Amazon VPC

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VPC Basics

- CIDR: Block of IP address
  - Example: 192.168.0.0/26: 192.168.0.0 – 192.168.0.63 (64 IP)
  - Used for security groups, route tables, VPC, subnets, etc…
- Private IP
  - 10.0.0.0 – 10.255.255.255 (10.0.0.0/8) <= in big networks
  - 172.16.0.0 – 172.31.255.255 (172.31.0.0/12)
  - 192.168.0.0 – 192.168.255.255 (192.168.0.0/16) <= example: home networks
- Public IP
  - All the rest

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VPC Basics

- VPC
  - A VPC must have a defined list of CIDR blocks, that cannot be changed
  - Each CIDR within VPC: min size is /28, max size is /16 (65536 IP addresses)
  - VPC is private, so only Private IP CIDR ranges are allowed
- Subnets
  - Within a VPC, defined as a CIDR that is a subset of the VPC CIDR
  - All instances within subnets get a private IP
  - First 4 IP and last one in every subnet is reserved by AWS
- Route Tables
  - Used to control where the network traffic is directed to
  - Can be associated with specific subnets
  - The "most specific" routing rule is always followed (192.168.0.1/24 beats 0.0.0.0/0)

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VPC Basics

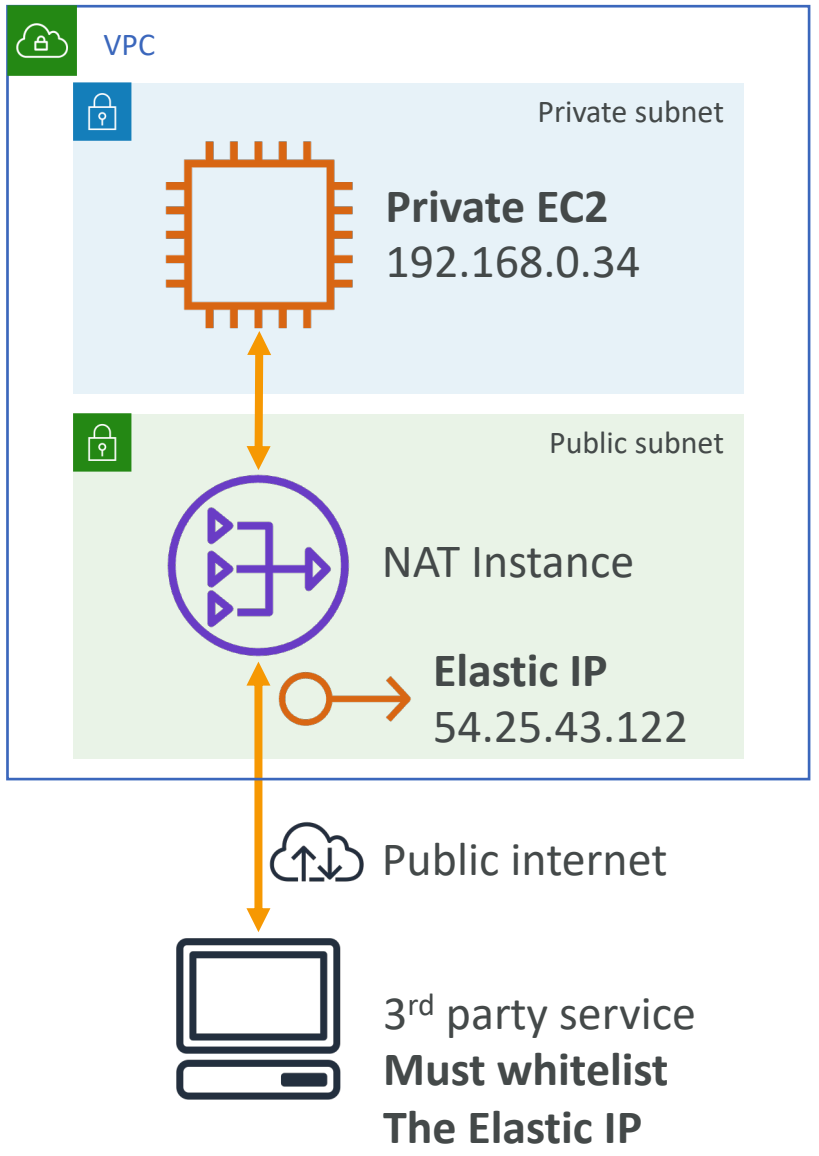
- Internet Gateway (IGW)
  - Helps our VPC connect to the internet, HA, scales horizontally
  - Acts as a NAT for instances that have a public IPv4 or public IPv6
- Public Subnets
  - Has a route table that sends 0.0.0.0/0 to an IGW
  - Instances must have a public IPv4 to talk to the internet
- Private Subnets
  - Access internet with a NAT Instance or NAT Gateway setup in a public subnet
  - Must edit routes so that 0.0.0.0/0 routes traffic to the NAT

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VPC Basics - NAT Instance

- EC2 instance you deploy in a public subnet
- Edit the route in your private subnet to route 0.0.0.0/0 to your NAT instance
- Not resilient to failure, limited bandwidth based on instance type, cheap
- Must manage failover yourself
- Must disable Source/Destination Check (EC2 setting)

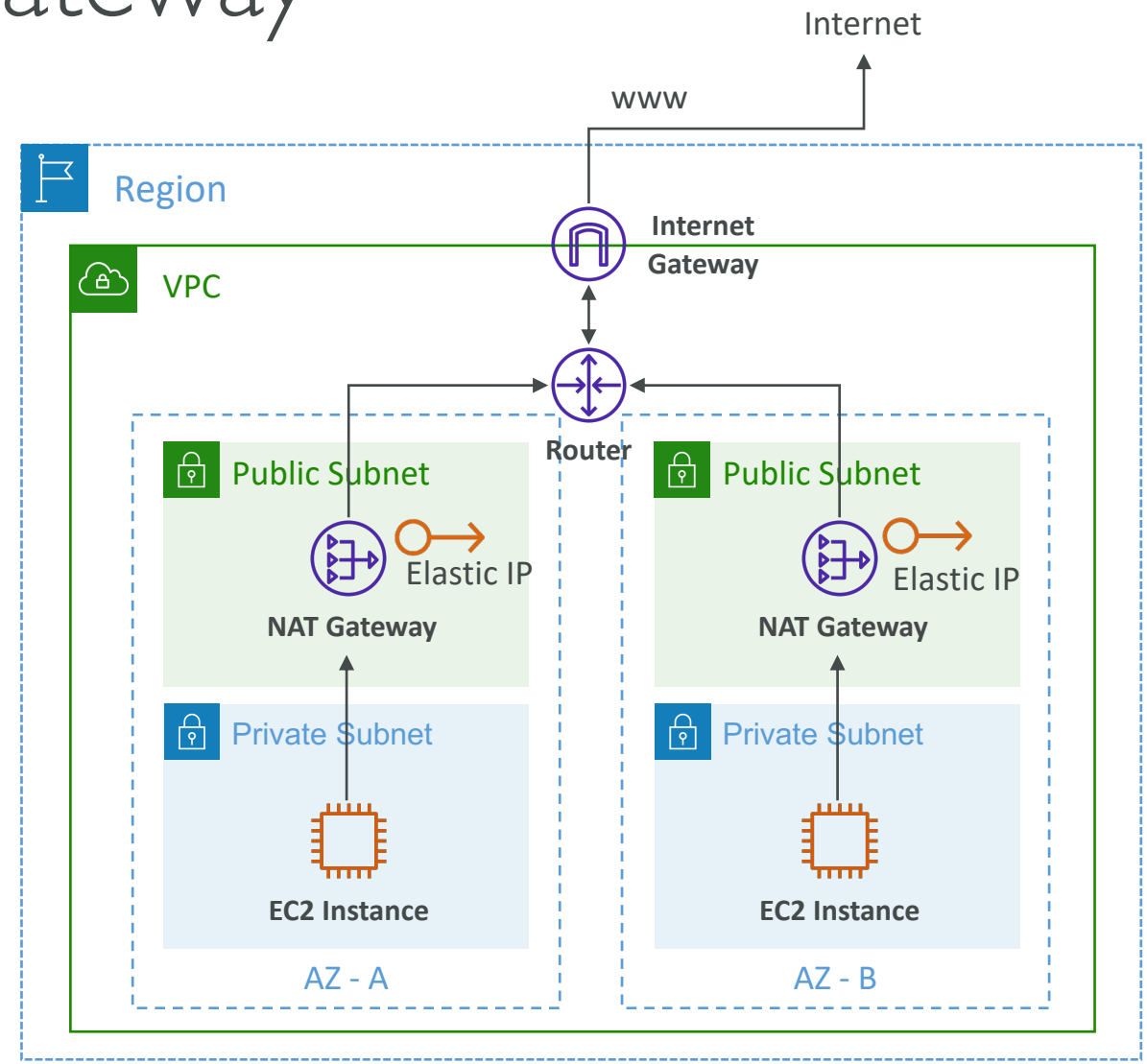


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VPC Basics – NAT Gateway

- Managed NAT solution, bandwidth scales automatically
- Resilient to failure within a single AZ
- Must deploy multiple NAT Gateways in multiple AZ for HA
- Has an Elastic IP, external services see the IP of the NAT Gateway as the source



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VPC Basics

- Network ACL (NACL)
  - Stateless firewall defined at the subnet level, applies to all instances within
  - Support for allow and deny rules
  - Stateless = return traffic must be explicitly allowed by rules
  - Helpful to quickly and cheaply block specific IP addresses

- Security Groups
  - Applied at the instance level, only support for allow rules, no deny rules
  - Stateful = return traffic is automatically allowed, regardless of rules
  - Can reference other security groups in the same region (peered VPC, cross-account)

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VPC Basics

- VPC Flows Logs
  - Log internet traffic going through your VPC
  - Can be defined at the VPC level, Subnet level, or ENI-level
  - Helpful to capture "denied internet traffic"
  - Can be sent to CloudWatch Logs and Amazon S3

- Bastion Hosts
  - SSH into private EC2 instances through a public EC2 instance (bastion host)
  - You must manage these instances yourself (failover, recovery)
  - SSM Session Manager is a more secure way to remote control without SSH

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VPC Basics

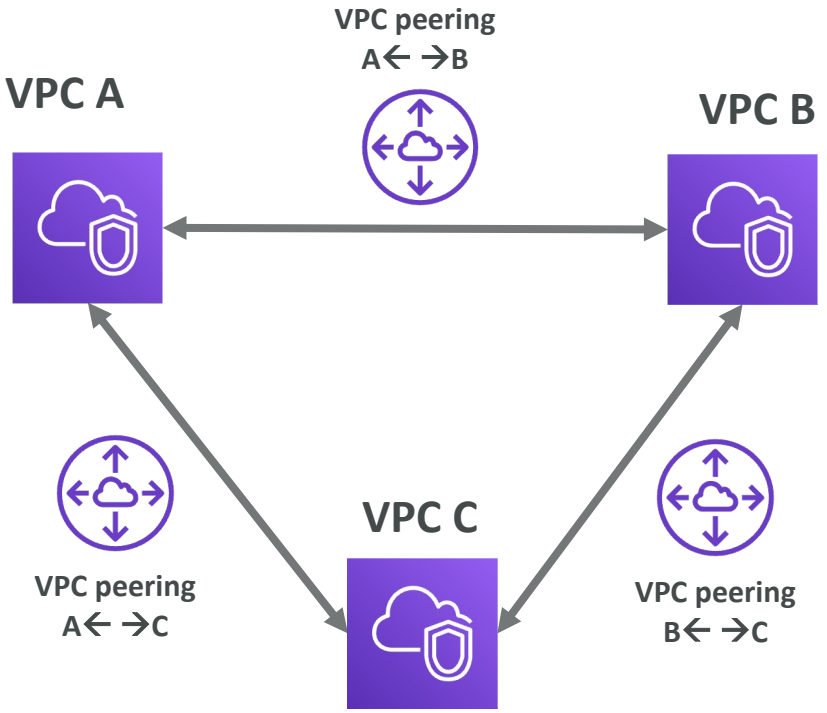
- IPv6 in short
  - All IPv6 addresses are public, total 3.4×10^{38} addresses (vs 4.3 billion IPv4)
  - Example CIDR: 2600:1f18:80c:a900::/56
  - Addresses are "random" and can't be scanned online (because too many)
- VPC support for IPv6
  - Create an IPv6 CIDR for VPC & use an IGW (supports IPv6)
  - Public subnet:
    - Create an instance with IPv6 support
    - Create a route table entry to ::/0 (IPv6 "all") to the IGW
  - Private subnet (instances cannot be reached by IPv6 but can reach IPv6):
    - Create an **Egress-Only Internet Gateway** in the VPC
    - Add a route table entry for the private subnet from ::/0 to the Egress-Only IGW

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VPC Peering

- Connect two VPC, privately using AWS' network
- Make them behave as if they were in the same network
- Must not have overlapping CIDR
- VPC Peering connection is **not transitive** (must be established for each VPC that need to communicate with one another)
- You can do VPC peering with another AWS account
- **You must update route tables in each VPC's subnets to ensure instances can communicate**



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VPC Peering – Good to know

- VPC peering can work **inter-region, cross-account**
- You can reference a security group of a peered VPC (works cross account)

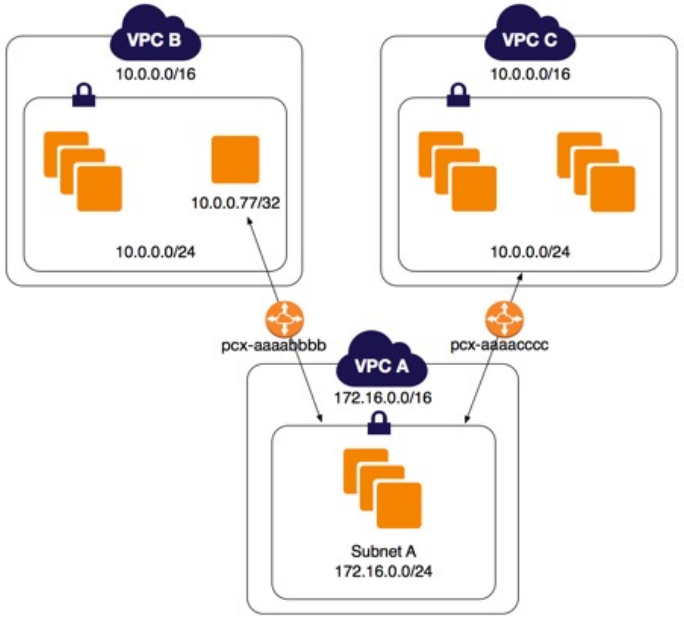
Type	Protocol	Port Range	Source
HTTP	TCP	80	sg-00d2b0f5fd6de757e
HTTP	TCP	80	sg-013347765f7a63aae/12356788

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VPC Peering – Longest Prefix Match

- VPC uses the longest prefix match to select the most specific route



Route Table	Destination	Target
VPC A	172.16.0.0/16	Local
	10.0.0.77/32	pcx-aaaabbbb
	10.0.0.0/16	pcx-aaaacccc
VPC B	10.0.0.0/16	Local
	172.16.0.0/16	pcx-aaaabbbb
VPC C	10.0.0.0/16	Local
	172.16.0.0/16	pcx-aaaacccc

- Here the longest prefix for 10.0.0.77 is **10.0.0.77/32** (route table VPC A)
- (other way of saying it is "most specific route")

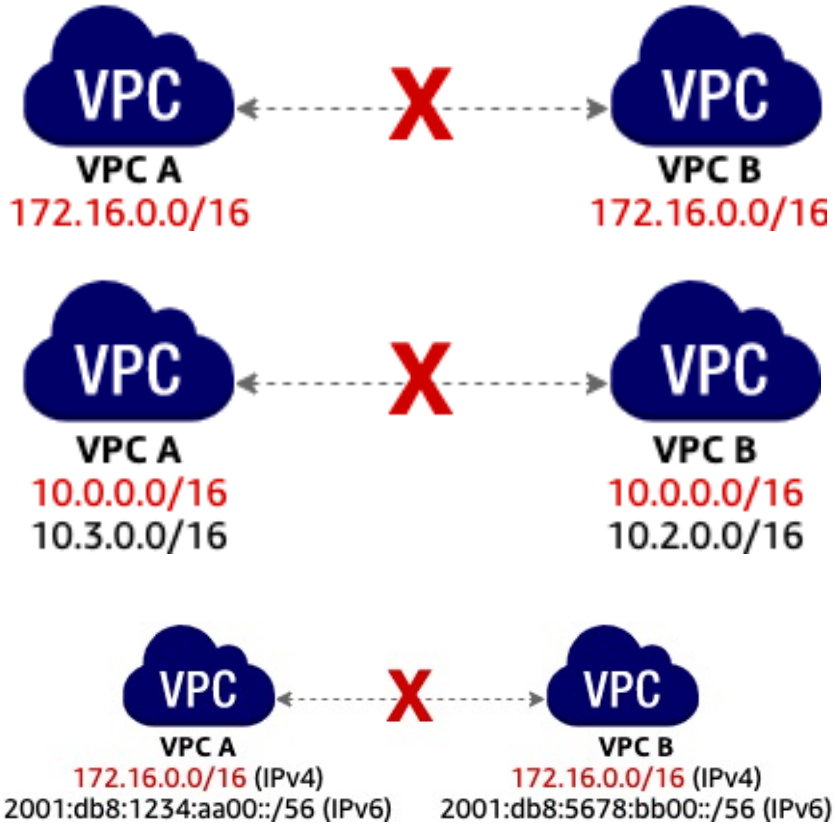
https://docs.aws.amazon.com/vpc/latest/peering/peering-configurations-partial-access.html#one-to-two-vpcs-lpm

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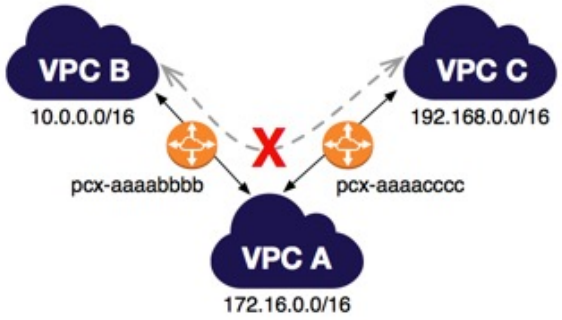
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VPC Peering – Invalid Configurations

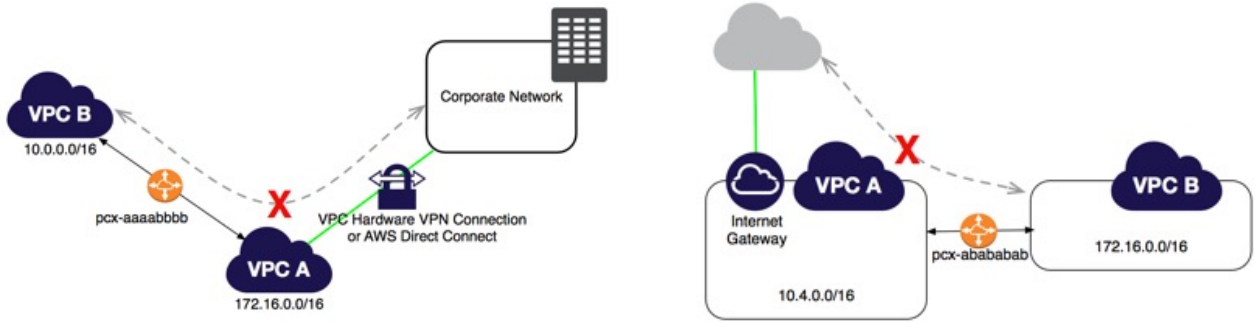
Overlapping CIDR for IPv4



No Transitive VPC Peering



No Edge to Edge Routing



VPN, Direct Connect, IGW, NAT, Gateway VPC Endpoint (S3 & DynamoDB)

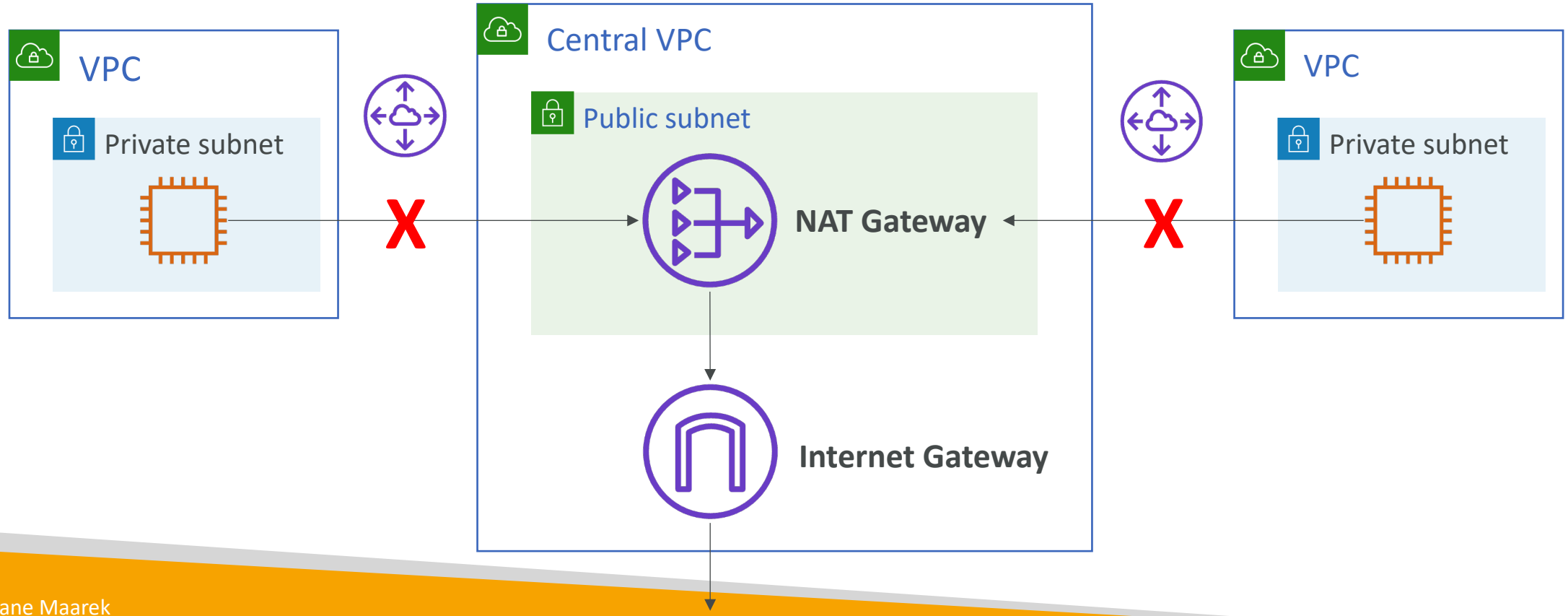
https://docs.aws.amazon.com/vpc/latest/peering/invalid-peering-configurations.html

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VPC Peering – Invalid Configuration
No edge to edge routing

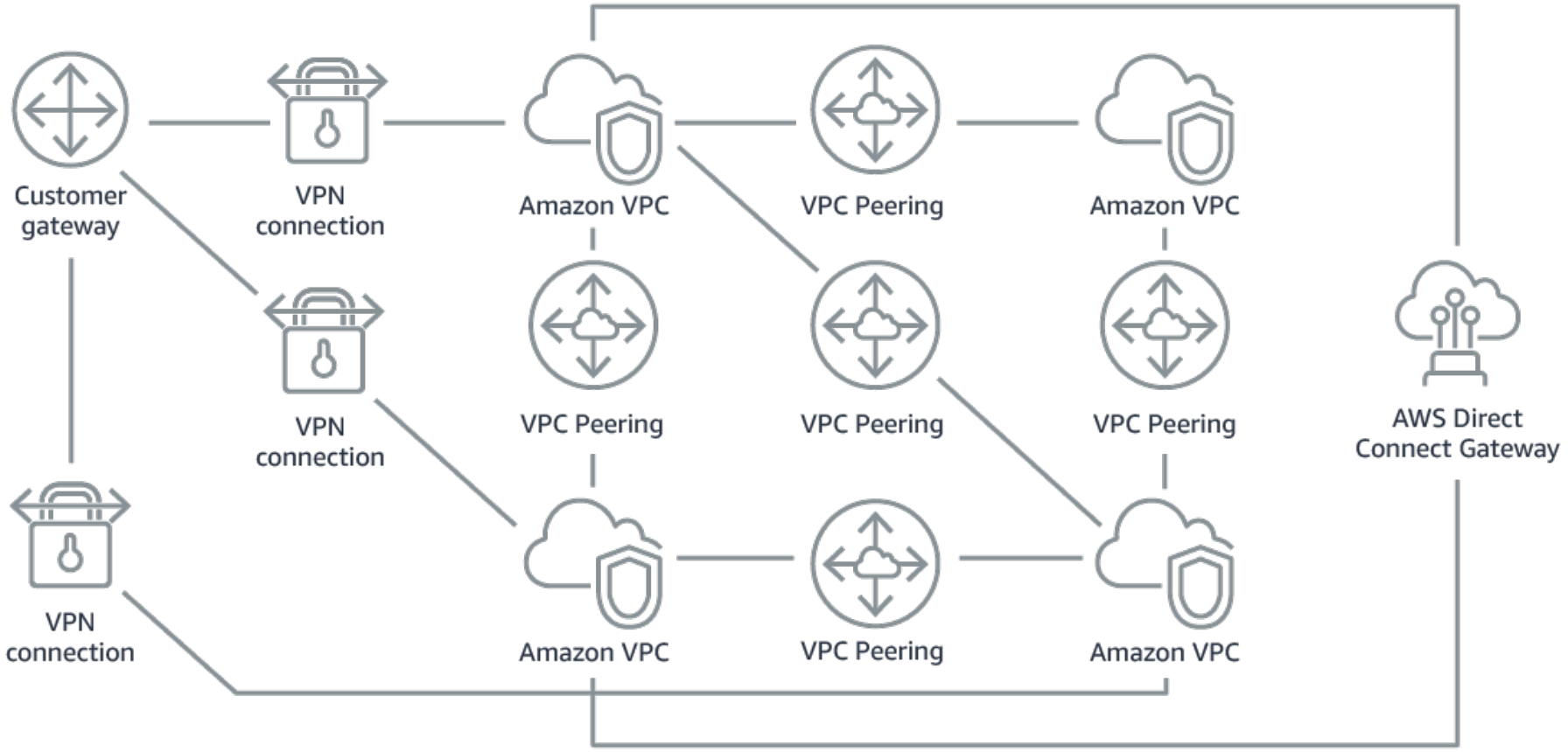
- This is an invalid configuration
- VPC Peering does not support edge to edge routing for NAT devices



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Network topologies can become complicated



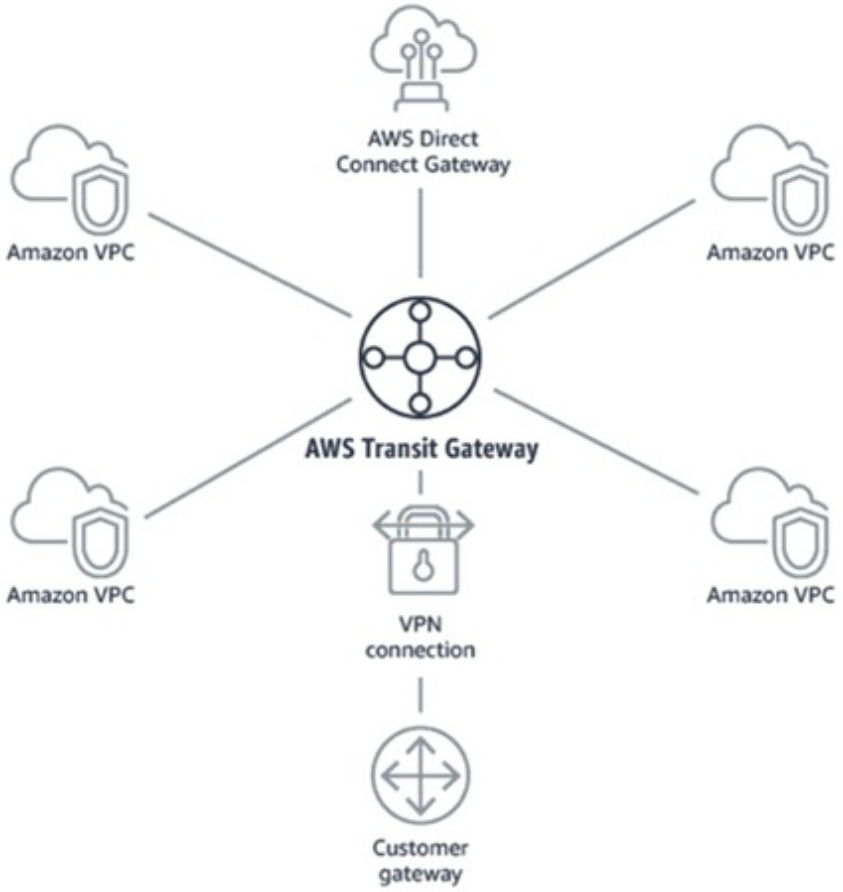
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Transit Gateway



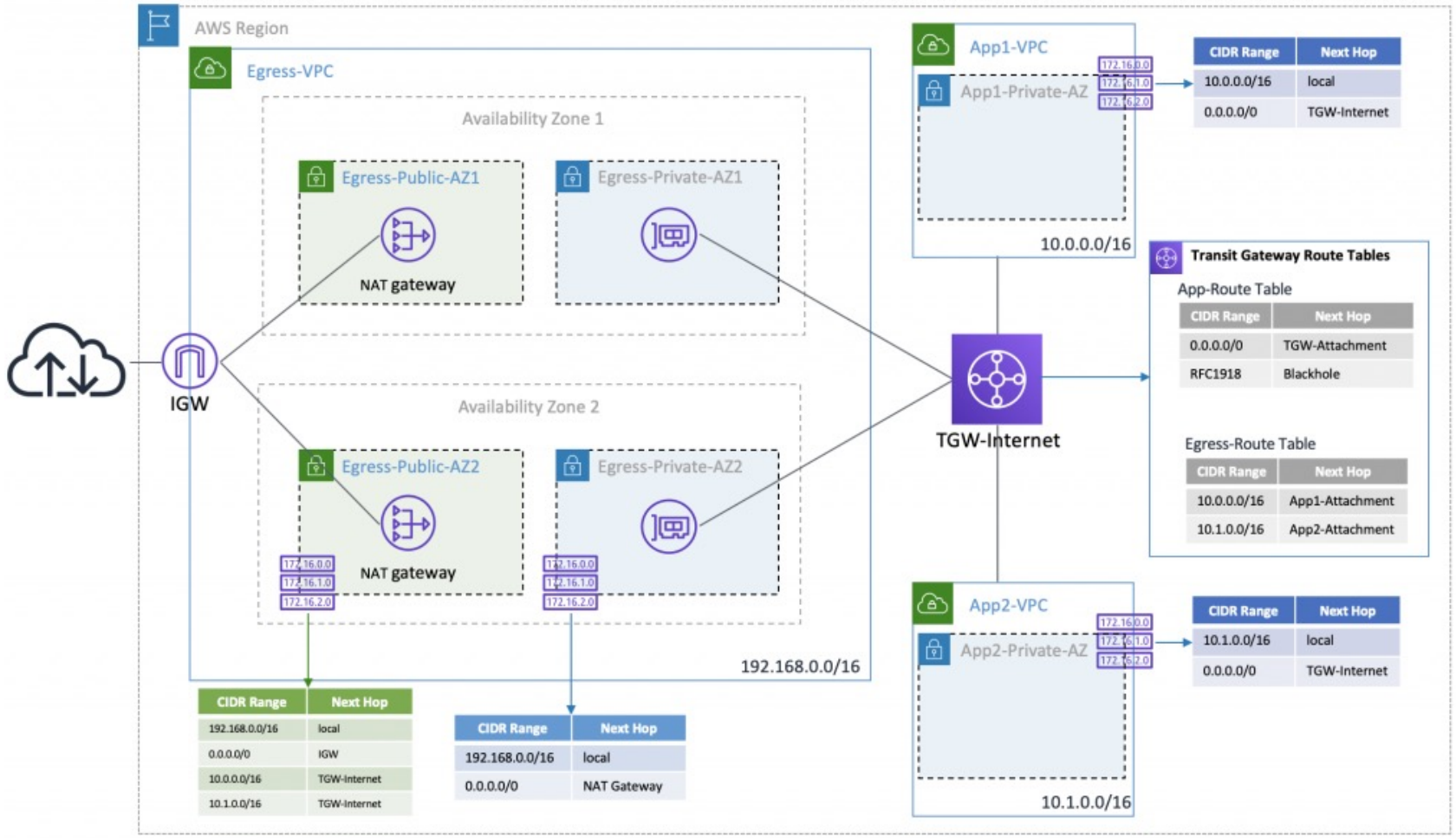
- For having transitive peering between thousands of VPC and on-premises, hub-and-spoke (star) connection
- Regional resource, can work cross-region
- Share cross-account using Resource Access Manager (RAM)
- You can peer Transit Gateways across regions
- Route Tables: limit which VPC can talk with other VPC
- Works with Direct Connect Gateway, VPN connections
- Supports **IP Multicast** (not supported by any other AWS service)
- Instances in a VPC can access a NAT Gateway, NLB, PrivateLink, and EFS in others VPCs attached to the AWS Transit Gateway.



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Transit Gateway – Central NAT Gateway



- The NAT Gateway is shared in the Egress-VPC
- The private App VPC can access internet through the TGW
- In this example: the App VPCs cannot communicate with each other based on the TGW route table

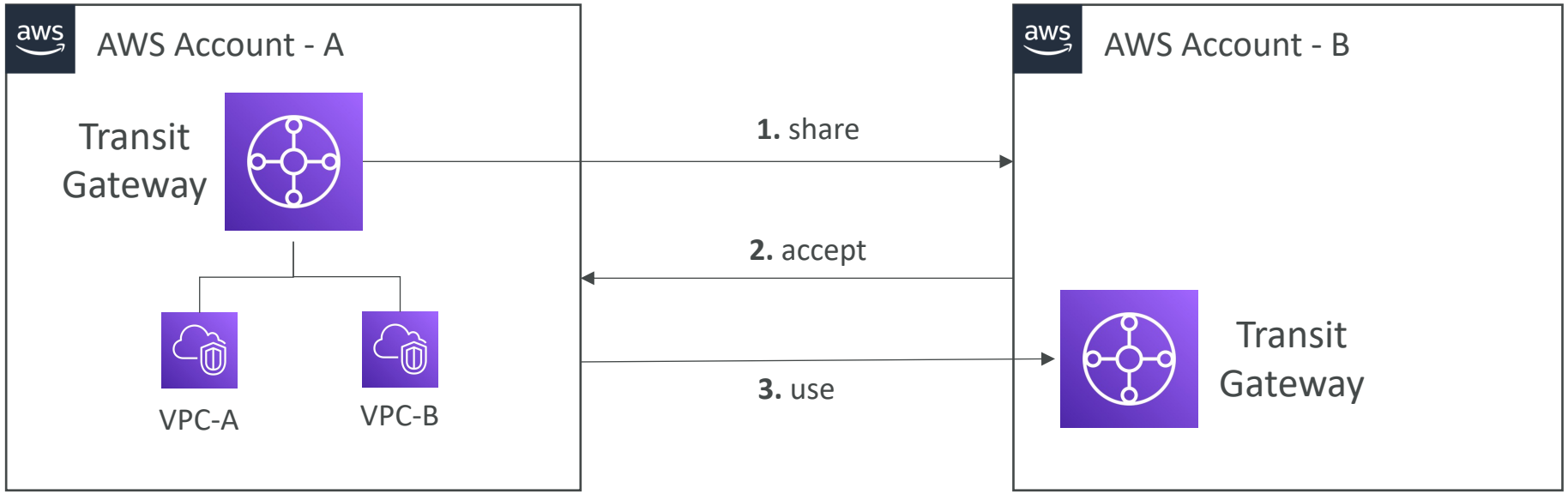
https://aws.amazon.com/blogs/networking-and-content-delivery/creating-a-single-internet-exit-point-from-multiple-vpcs-using-aws-transit-gateway/

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Transit Gateway – Sharing through RAM

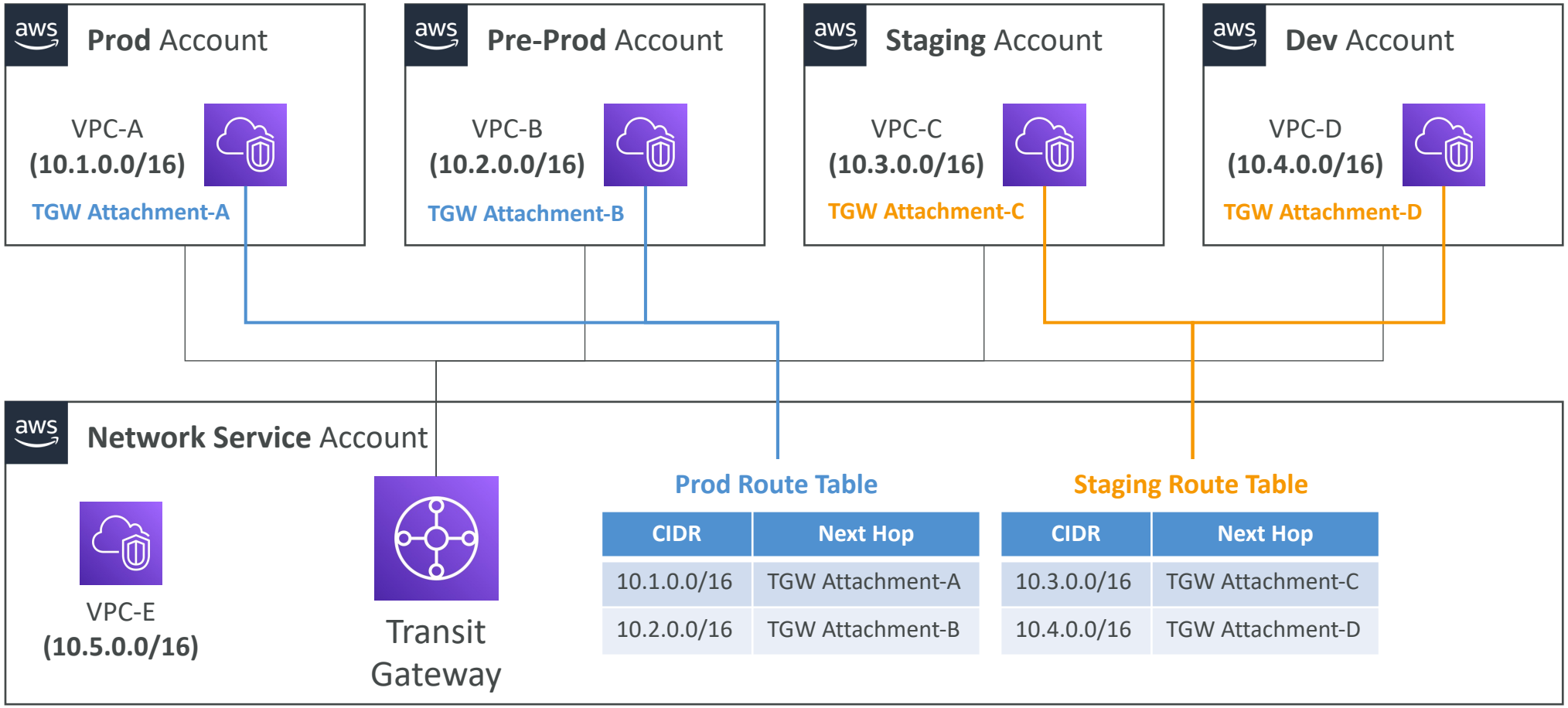
- You can use AWS RAM to share a Transit Gateway for VPC attachments across accounts or across AWS Organization



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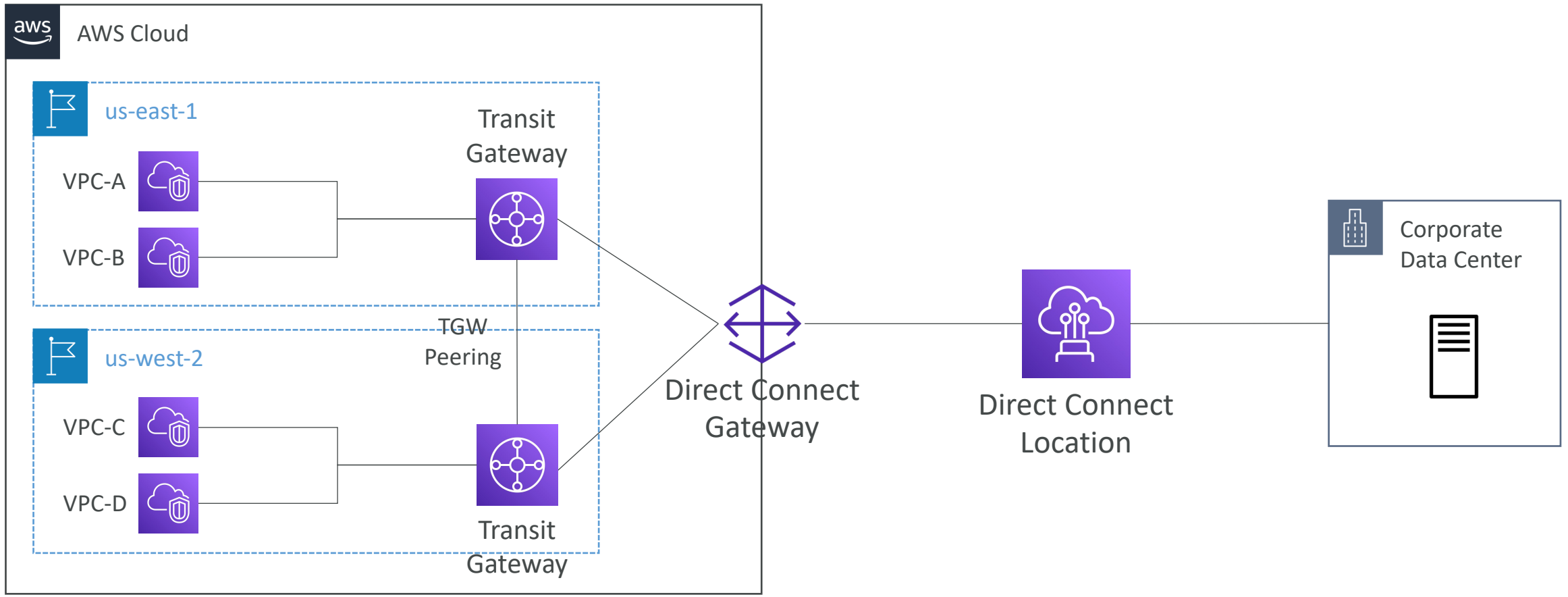
Transit Gateway – Use Different Route Tables to Prevent VPC from Communicating



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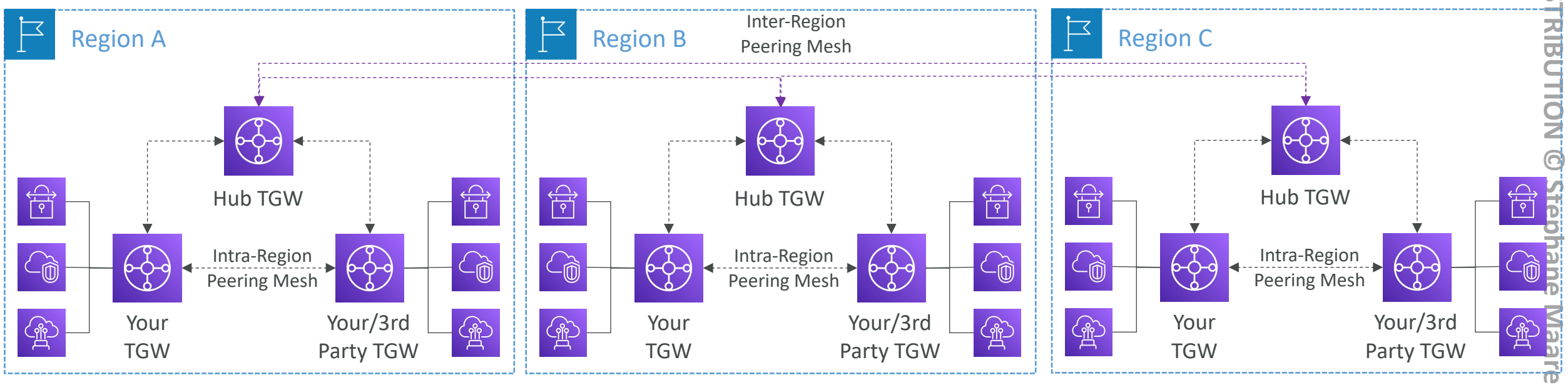
Transit Gateway to Direct Connect Gateway



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Transit Gateway – Inter & Intra Region Peering



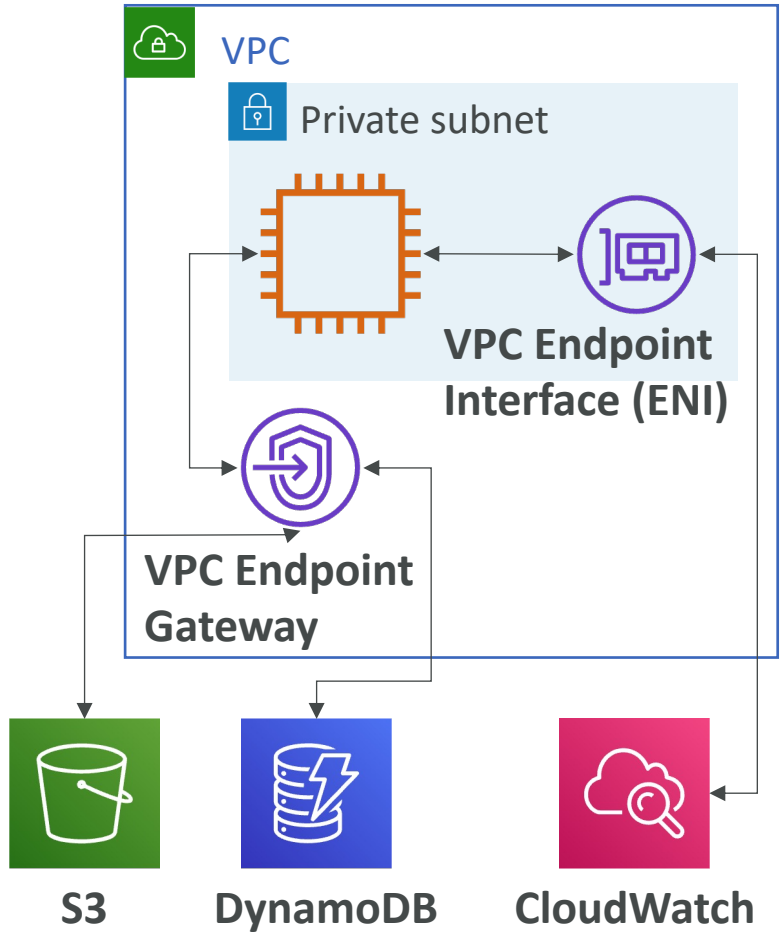
- Billed hourly for each peering attachment, no data processing charges
- If data goes cross-region through the attachment, standard charges
- Specify the Transit Gateway ID

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VPC Endpoints

- Endpoints allow you to connect to AWS Services using a private network instead of the public www network
- They scale horizontally and are redundant
- No more IGW, NAT, etc… to access AWS Services
- VPC Endpoint Gateway (S3 & DynamoDB)
- VPC Endpoint Interface (all services)
- In case of issues:
  - Check DNS Setting Resolution in your VPC
  - Check Route Tables
- Supports cross-region endpoints

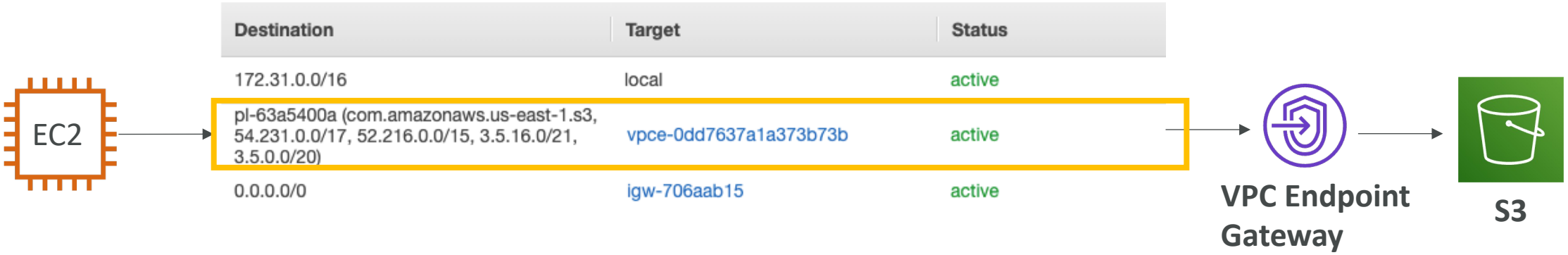


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VPC Endpoint Gateway

- Only works for S3 and DynamoDB, must create one gateway per VPC
- Must update route tables entries
- Gateway is defined at the VPC level



- DNS resolution must be enabled in the VPC
- The same public hostname for S3 can be used
- Gateway endpoint cannot be extended out of a VPC (VPN, DX, TGW, peering)

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VPC Endpoints Interface

- Provision an ENI that will have a private endpoint interface hostname
- Leverage Security Groups for security
- Private DNS (setting when you create the endpoint)
  - The public hostname of a service will resolve to the private Endpoint Interface hostname
  - VPC Setting: "Enable DNS hostnames" and "Enable DNS Support" must be 'true'
  - Example for Athena:
    - vpce-0b7d2995e9dfe5418-mwrths3x.athena.us-east-1.vpce.amazonaws.com
    - vpce-0b7d2995e9dfe5418-mwrths3x-us-east-1a.athena.us-east-1.vpce.amazonaws.com
    - vpce-0b7d2995e9dfe5418-mwrths3x-us-east-1b.athena.us-east-1.vpce.amazonaws.com
    - **athena.us-east-1.amazonaws.com** (private DNS name)
- Interface can be accessed from Direct Connect and Site-to-Site VPN

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VPC Endpoint Policies

- Endpoint Policies are JSON documents to control access to services
- Does not override or replace IAM user policies or service-specific policies (such as S3 bucket policies)

```
{
   "Statement": [{
      "Action": ["sqs:SendMessage"],
      "Effect": "Allow",
      "Resource": "arn:aws:sqs:us-east-2:123456789012:MyQueue",
      "Principal": {
        "AWS": "arn:aws:iam:123456789012:user/MyUser"
      }
   }]
}
```

- **Note:** the IAM user can still use other SQS API from outside the VPC Endpoint
- You could add an SQS queue policy to deny any action not done through the VPC endpoint

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VPC Endpoint Policy & S3 bucket policy

- VPC Endpoint Policy to restrict access to bucket "my_secure_bucket"

```
{
  "Statement": [
    {
      "Sid": "Access-to-specific-bucket-only",
      "Principal": "*",
      "Action": [
        "s3:GetObject",
        "s3:PutObject"
      ],
      "Effect": "Allow",
      "Resource": ["arn:aws:s3:::my_secure_bucket",
                   "arn:aws:s3:::my_secure_bucket/*"]
    }
  ]
}
```

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VPC Endpoint Policy & S3 bucket policy

- VPC Endpoint Policy to allow access to Amazon Linux 2 repositories

```
{
  "Statement": [
    {
      "Sid": "AmazonLinux2AMIRepositoryAccess",
      "Principal": "*",
      "Action": [
        "s3:GetObject"
      ],
      "Effect": "Allow",
      "Resource": [
        "arn:aws:s3:::amazonlinux.*.amazonaws.com/*"
      ]
    }
  ]
}
```

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VPC Endpoint Policy & S3 bucket policy

- S3 bucket policy may have
  - **Condition: "aws:sourceVpce": "vpce-1a2b3c4d"** to Deny any traffic that doesn't come from a specific VPC endpoint (more secure)
  - **Condition: "aws:sourceVpc": "vpc-111bbb22"** for a specific VPC
- The **aws:sourceVpc** condition only works for VPC Endpoints, in case you have multiple endpoints and want to manage access to your S3 buckets for all your endpoints
- The S3 bucket policies can restrict access only from a specific **public IP** address or an elastic IP address. You can't restrict based on private IP
- Therefore aws:SourceIp condition doesn't apply for VPC endpoints

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Example S3 bucket policies

- S3 bucket policy to restrict to one specific VPC Endpoint

```
{
  "Version": "2012-10-17",
  "Id": "Policy1415115909152",
  "Statement": [
    {
      "Sid": "Access-to-specific-VPCE-only",
      "Principal": "*",
      "Action": "s3:*",
      "Effect": "Deny",
      "Resource": ["arn:aws:s3:::my_secure_bucket",
                   "arn:aws:s3:::my_secure_bucket/*"],
      "Condition": {
        "StringNotEquals": {
          "aws:sourceVpce": "vpce-1a2b3c4d"
        }
      }
    }
  ]
}
```

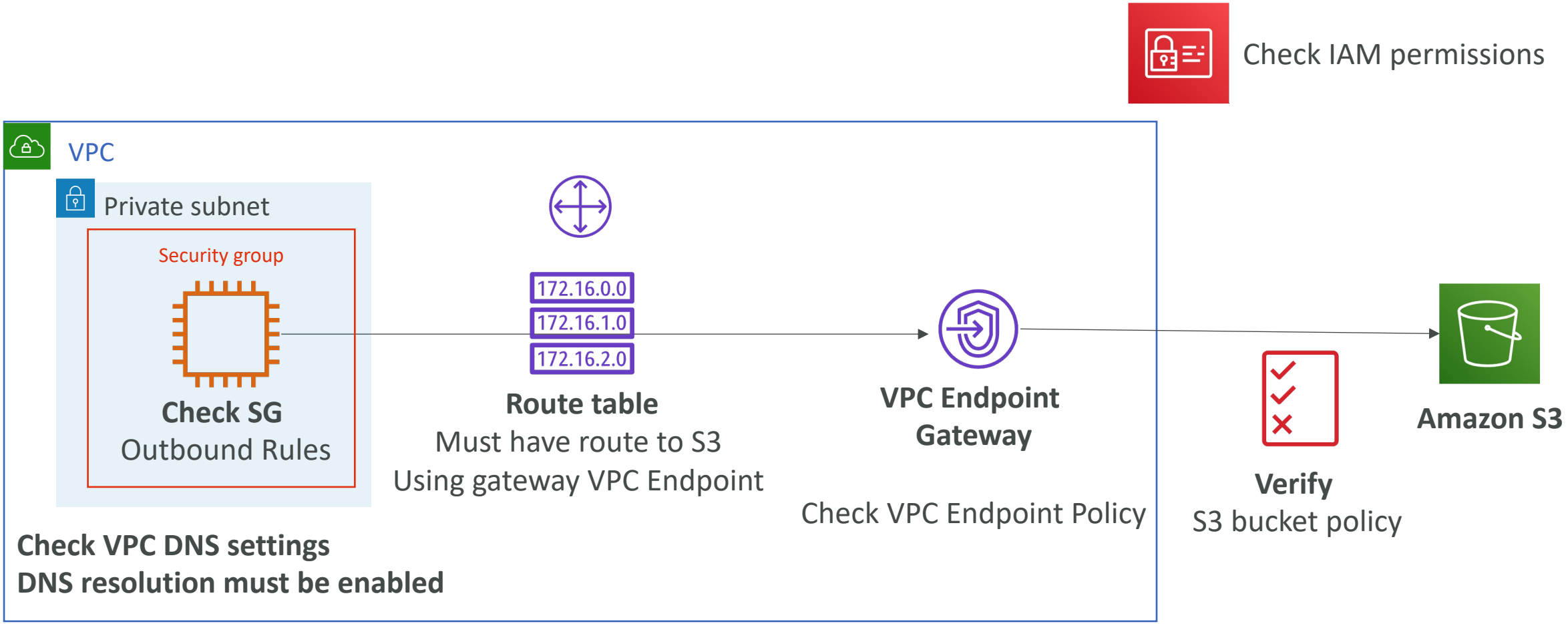
- S3 bucket policy to restrict to an entire VPC (multiple VPC Endpoints)

```
{
  "Version": "2012-10-17",
  "Id": "Policy1415115909152",
  "Statement": [
    {
      "Sid": "Access-to-specific-VPC-only",
      "Principal": "*",
      "Action": "s3:*",
      "Effect": "Deny",
      "Resource": ["arn:aws:s3:::my_secure_bucket",
                   "arn:aws:s3:::my_secure_bucket/*"],
      "Condition": {
        "StringNotEquals": {
          "aws:sourceVpc": "vpc-111bbb22"
        }
      }
    }
  ]
}
```

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VPC Endpoint Policies for S3 Troubleshooting

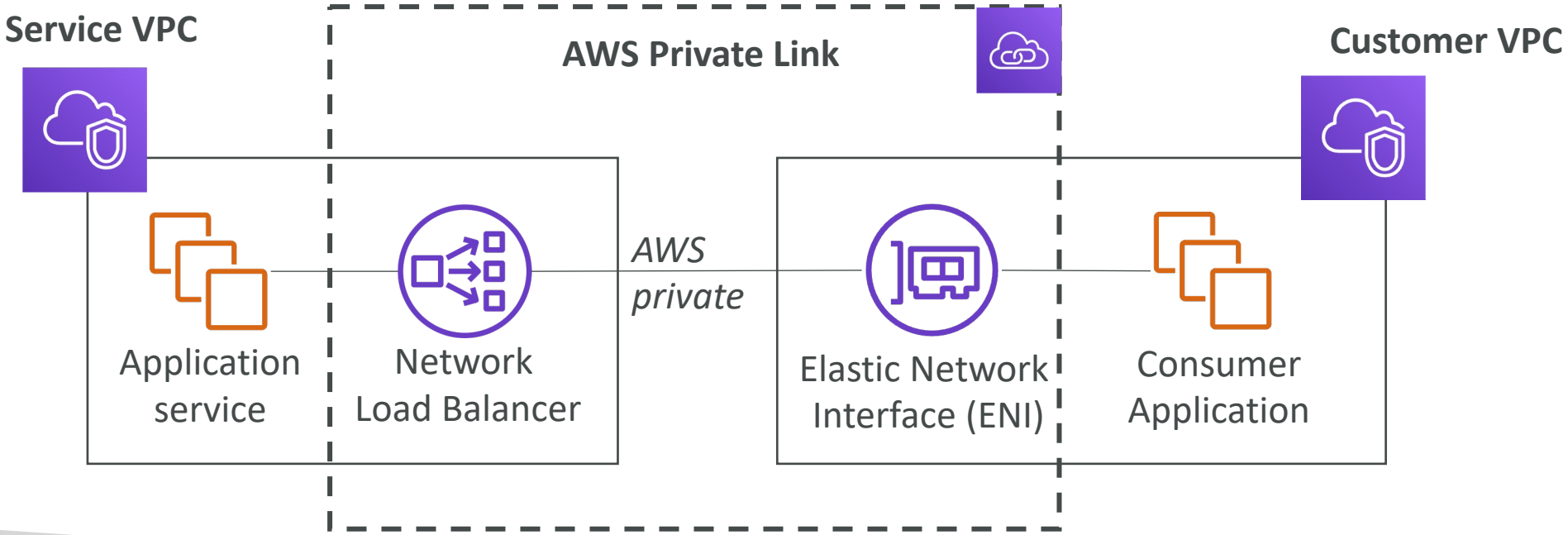


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AWS PrivateLink (VPC Endpoint Services)

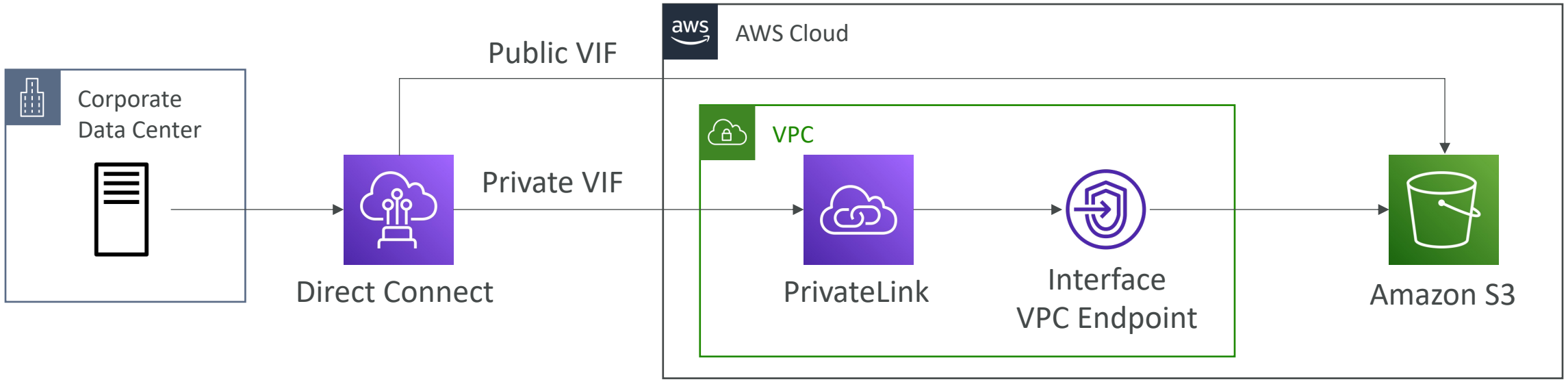
- Most secure & scalable way to expose a service to 1000s of VPC (own or other accounts)
- Does not require VPC peering, internet gateway, NAT, route tables…
- Requires a network load balancer (Service VPC) and ENI (Customer VPC)
- If the NLB is in multiple AZ, and the ENI in multiple AZ, the solution is fault tolerant!



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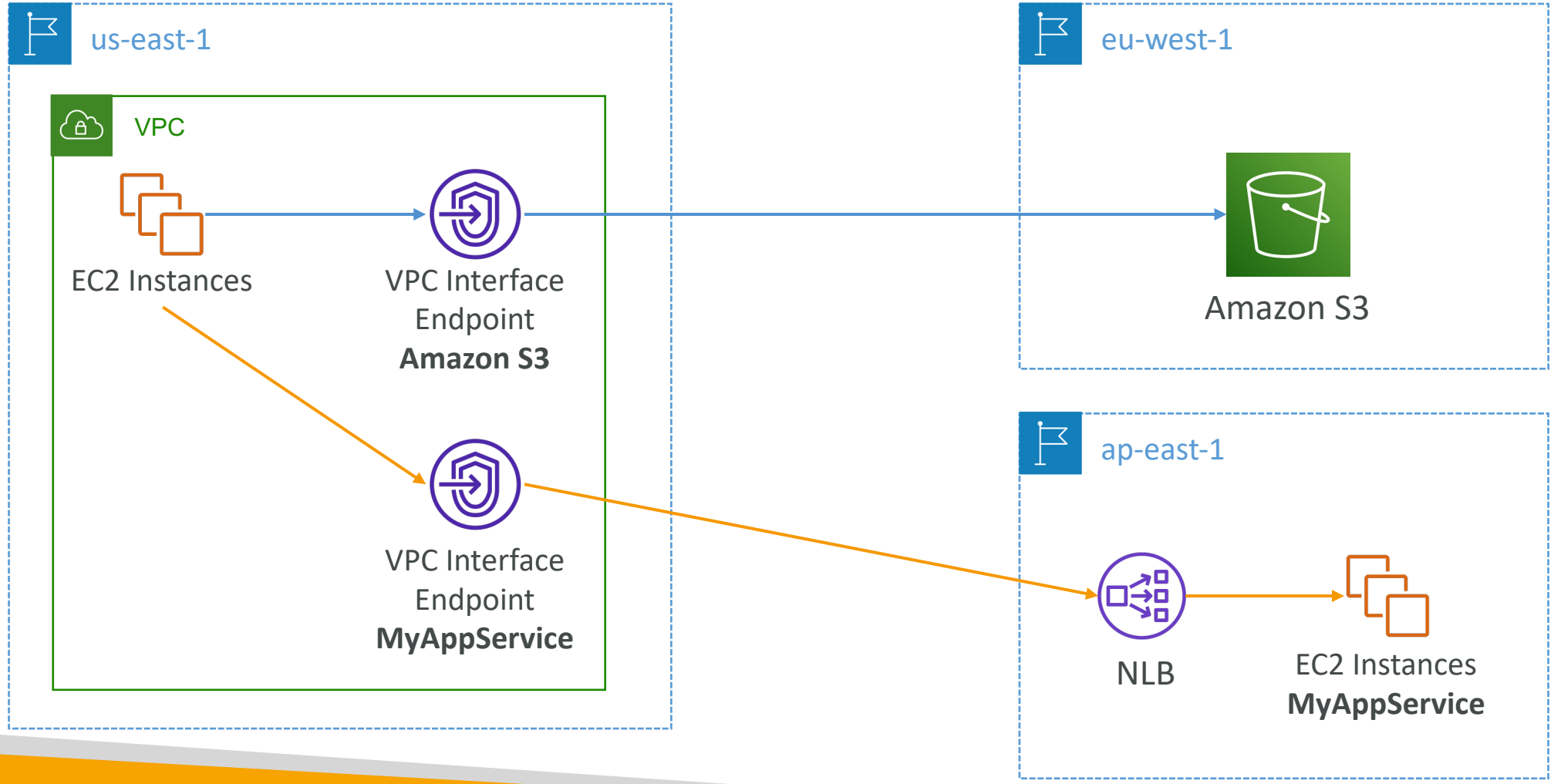
PrivateLink for Amazon S3 with Direct Connect



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VPC Endpoints / PrivateLink Cross-Region Connectivity

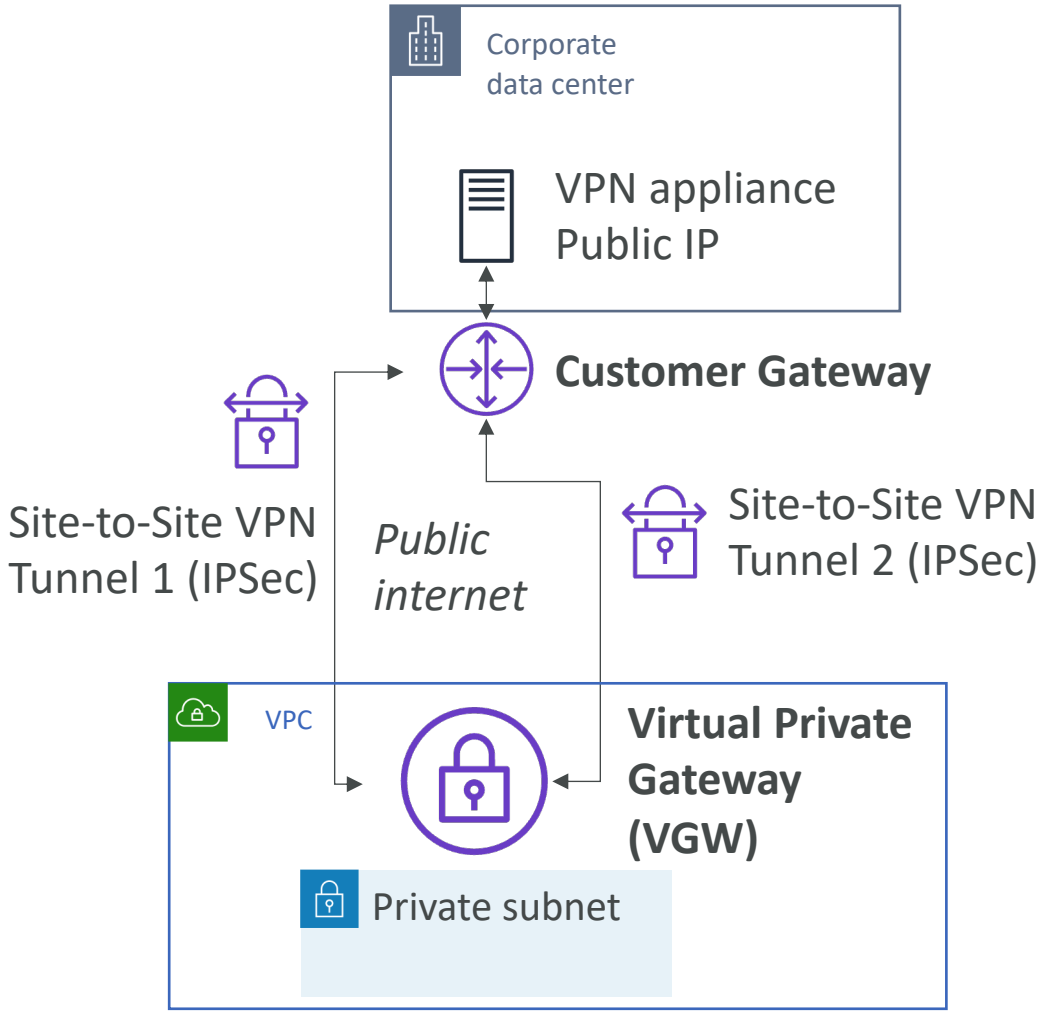


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Site to Site VPN (AWS Managed VPN)

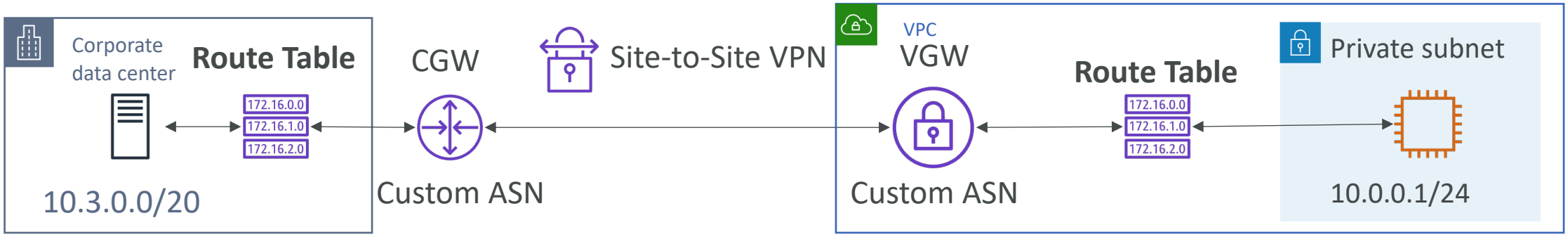
- on-premises:
  - Setup a software or hardware VPN appliance to your on-premises network.
  - The on-premises VPN should be accessible using a public IP
- AWS-side:
  - Setup a Virtual Private Gateway (VGW) and attach to your VPC
  - Setup a Customer Gateway to point the on-premises VPN appliance
- Two VPN connections (tunnels) are created for redundancy, encrypted using IPSec
- Can optionally accelerate it using Global Accelerator (for worldwide networks)



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Route Propagation in Site-to-Site VPN



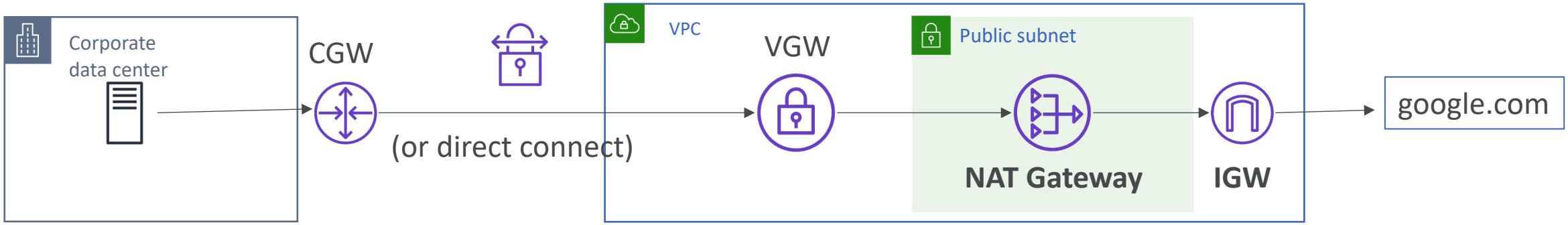
- Static Routing:
  - Create static route in corporate data center for 10.0.0.1/24 through the CGW
  - Create static route in AWS for 10.3.0.0/20 through the VGW
- Dynamic Routing (BGP):
  - Uses BGP (Border Gateway Protocol) to share routes automatically (eBGP for internet)
  - We don't need to update the routing tables, it will be done for us dynamically
  - Just need to specify the ASN (Autonomous System Number) of the CGW and VGW

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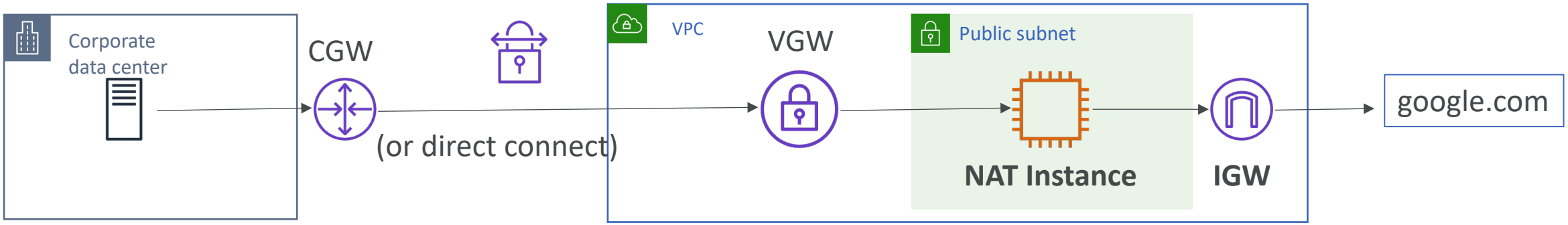
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Site to Site VPN and Internet Access

- NOT OKAY (blocked by NAT Gateway restrictions)



- OKAY (self managed NAT Instance – more control)

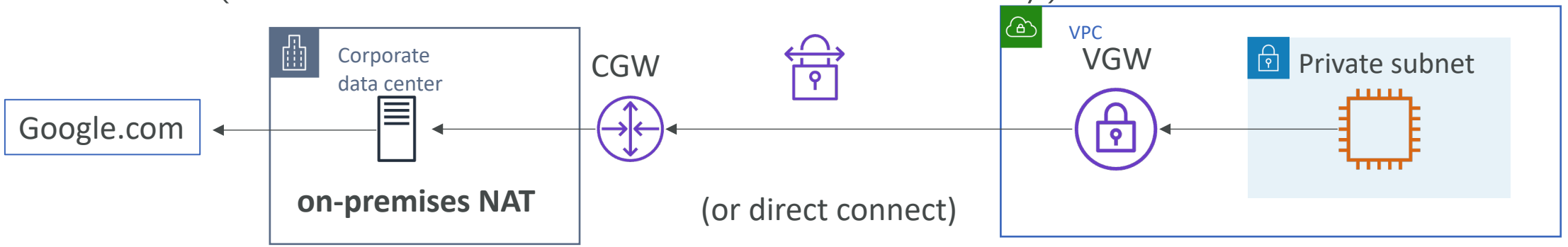


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Site to Site VPN and Internet Access

- OKAY (alternative to NAT Instances / Gateway)

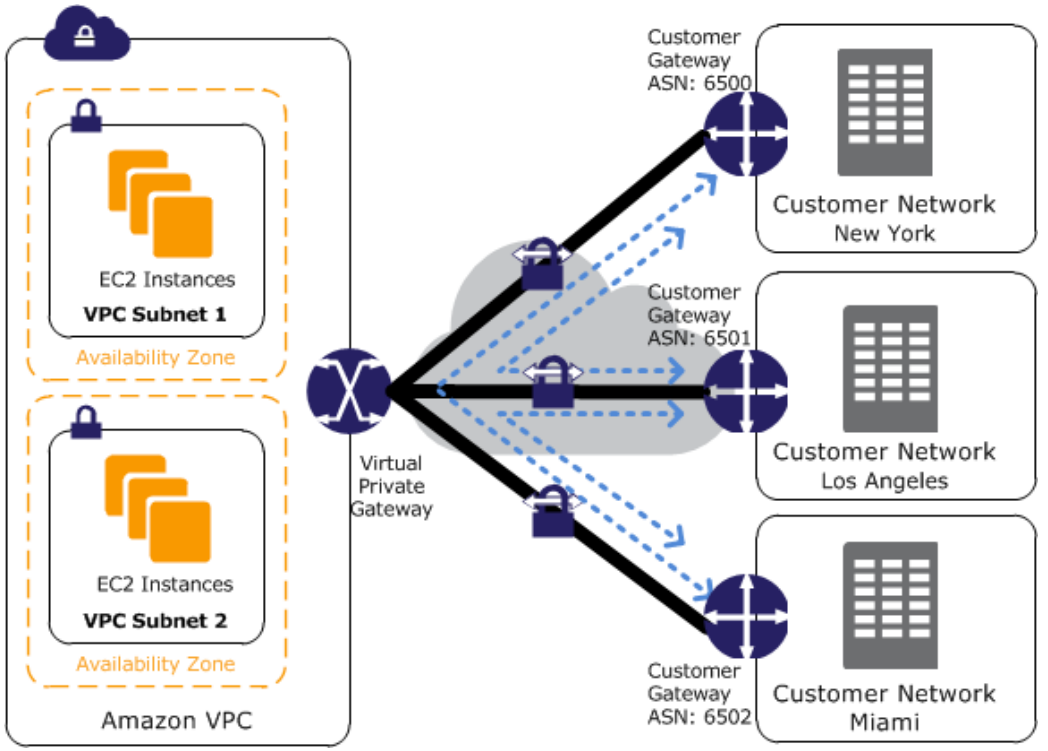


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AWS VPN CloudHub

- Can connect up to 10 Customer Gateway for each Virtual Private Gateway (VGW)
- Low cost hub-and-spoke model for primary or secondary network connectivity between locations
- Provide secure communication between sites, if you have multiple VPN connections
- It's a VPN connection so it goes over the public internet
- Can be a **failover connection** between your on-premises locations

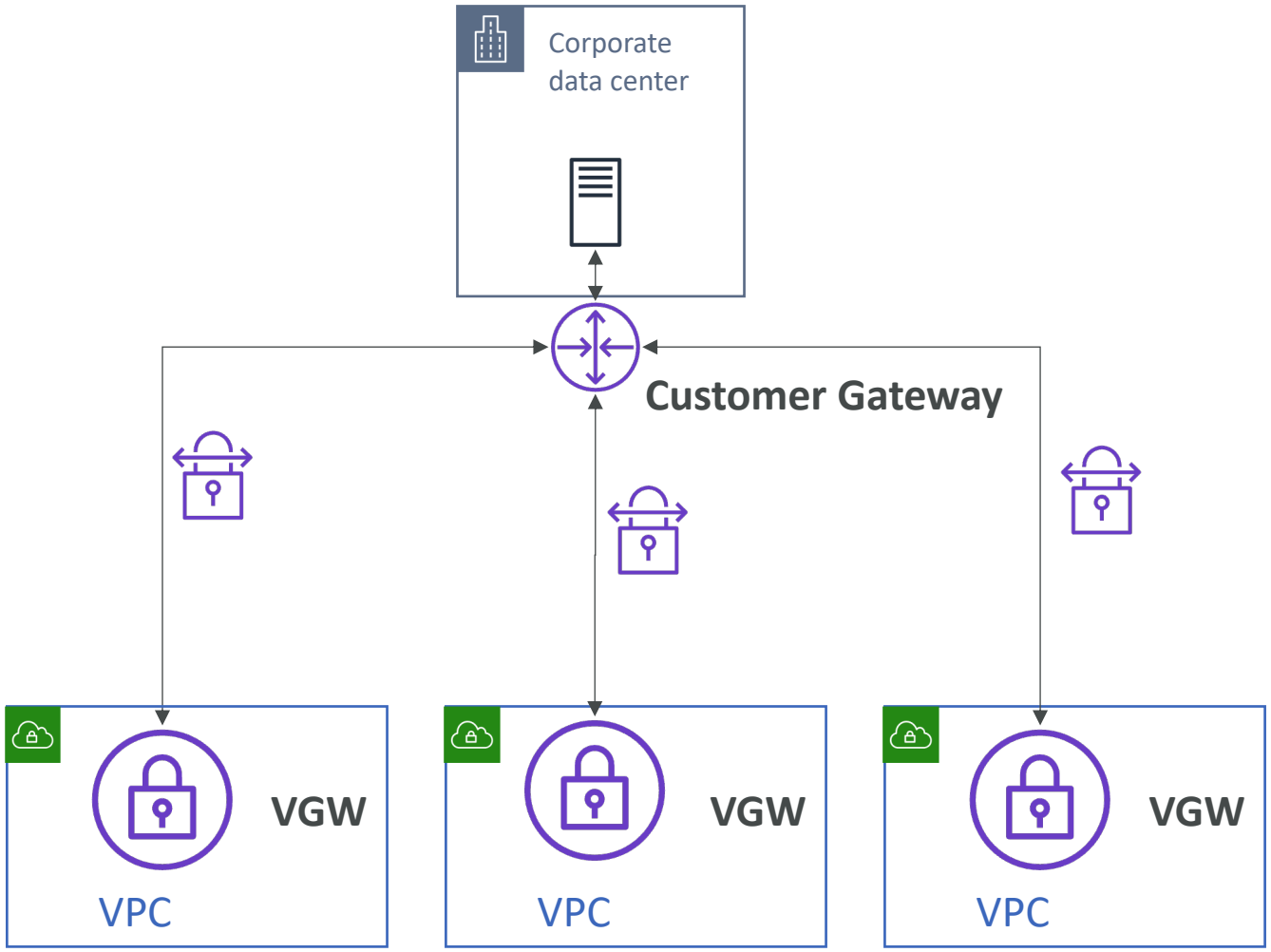


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VPN to multiple VPC

- For VPN-based customers, AWS recommends creating a separate VPN connection for each customer VPC.
- Direct Connect is recommended because it has a Direct Connect Gateway

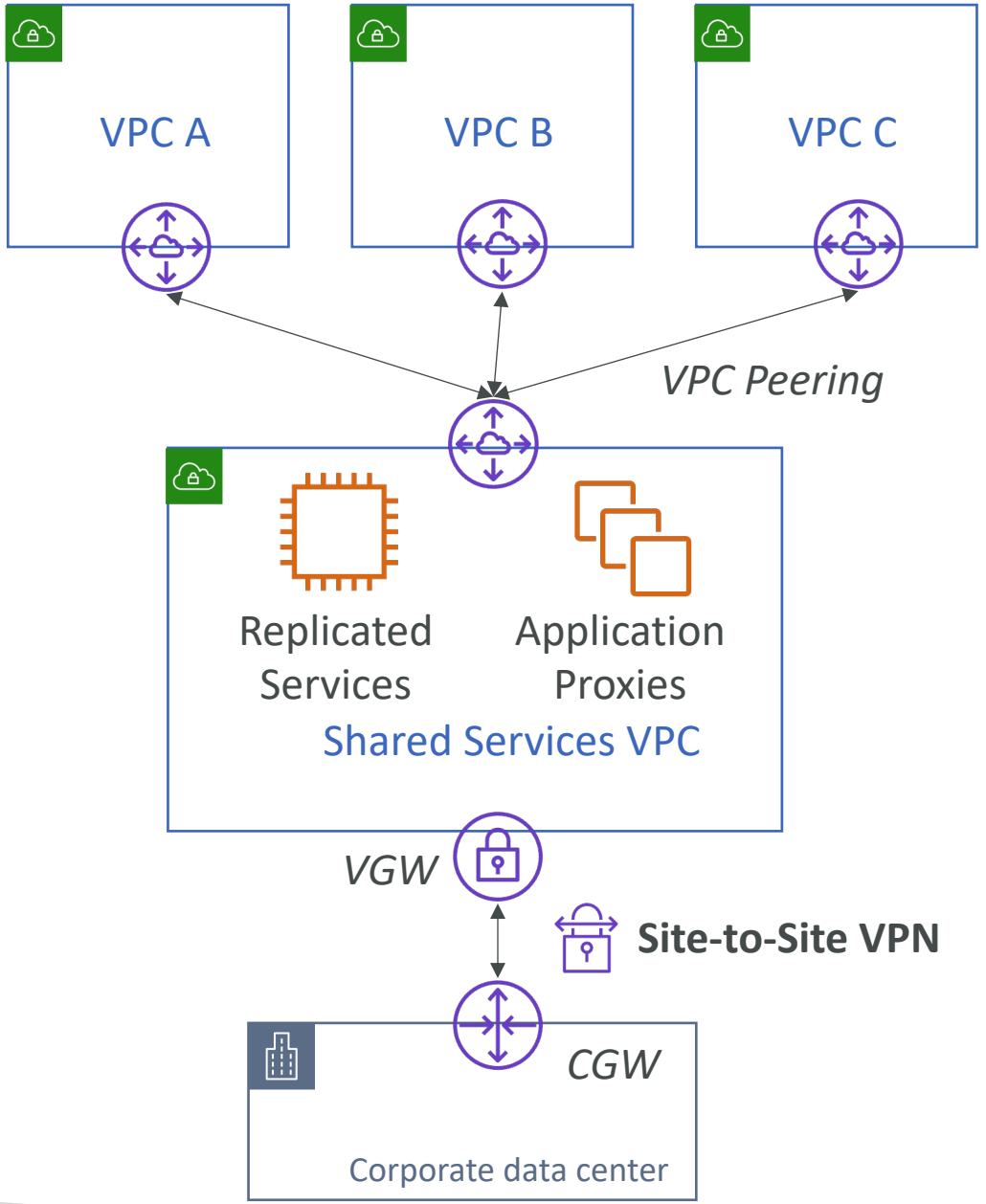


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Shared Services VPC

- Create a VPN connection between on-premises and shared service VPC
- Replicate services, applications, databases between on-premises and the Shared Services VPC or deploy proxies in the shared service VPC
- Do VPC peering between the VPC and the shared service VPC
- VPCs can directly access the Shared Service VPC services and do not need VPN connections to on-premises

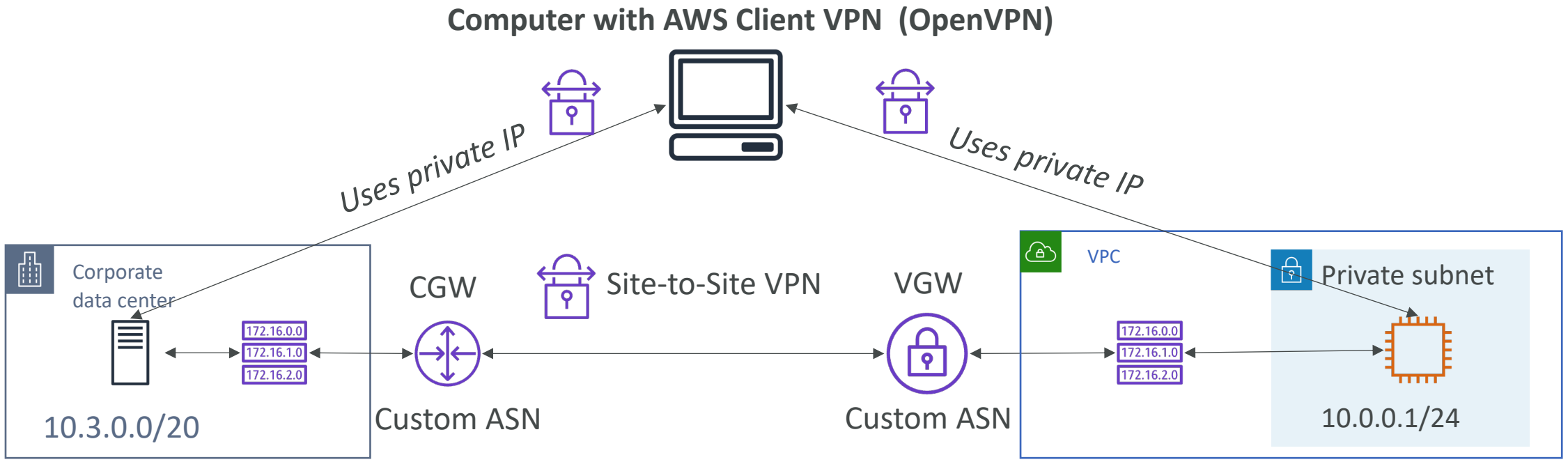


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AWS Client VPN

- Connect from your computer using OpenVPN to your private network in AWS and on-premises

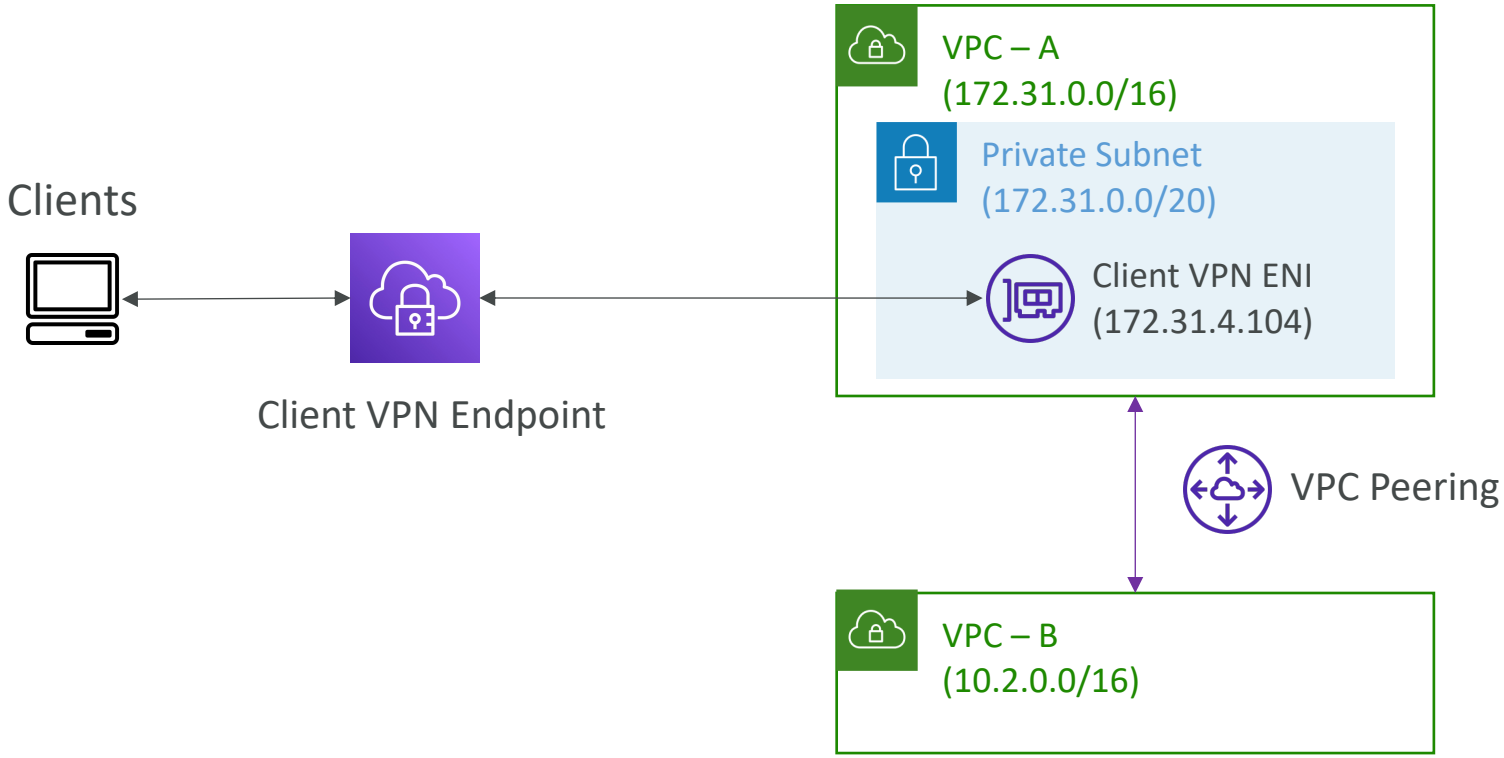


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Client VPN – Peered VPC

- Client VPN is compatible with VPC peering

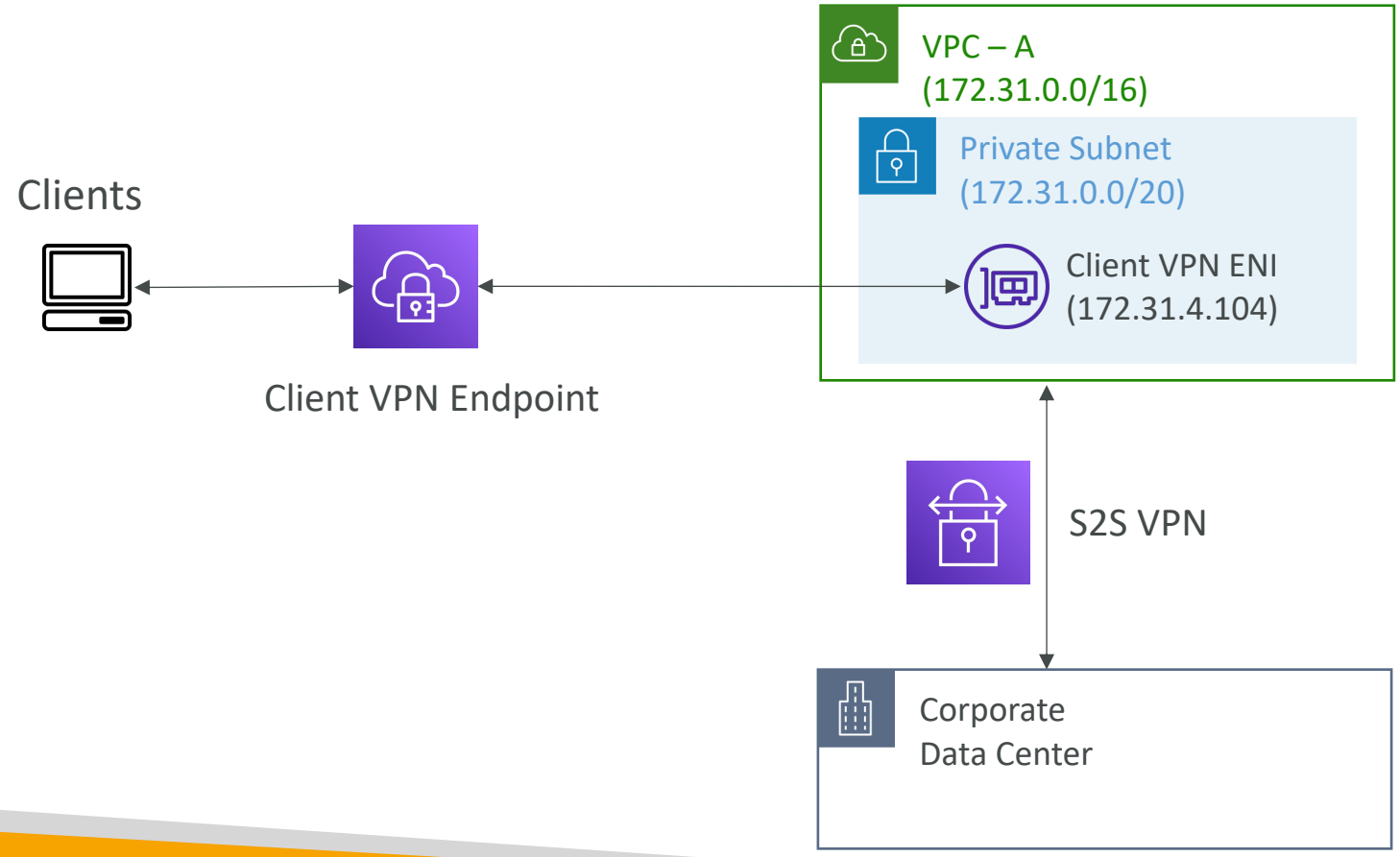


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Client VPN – Access to On-Premises

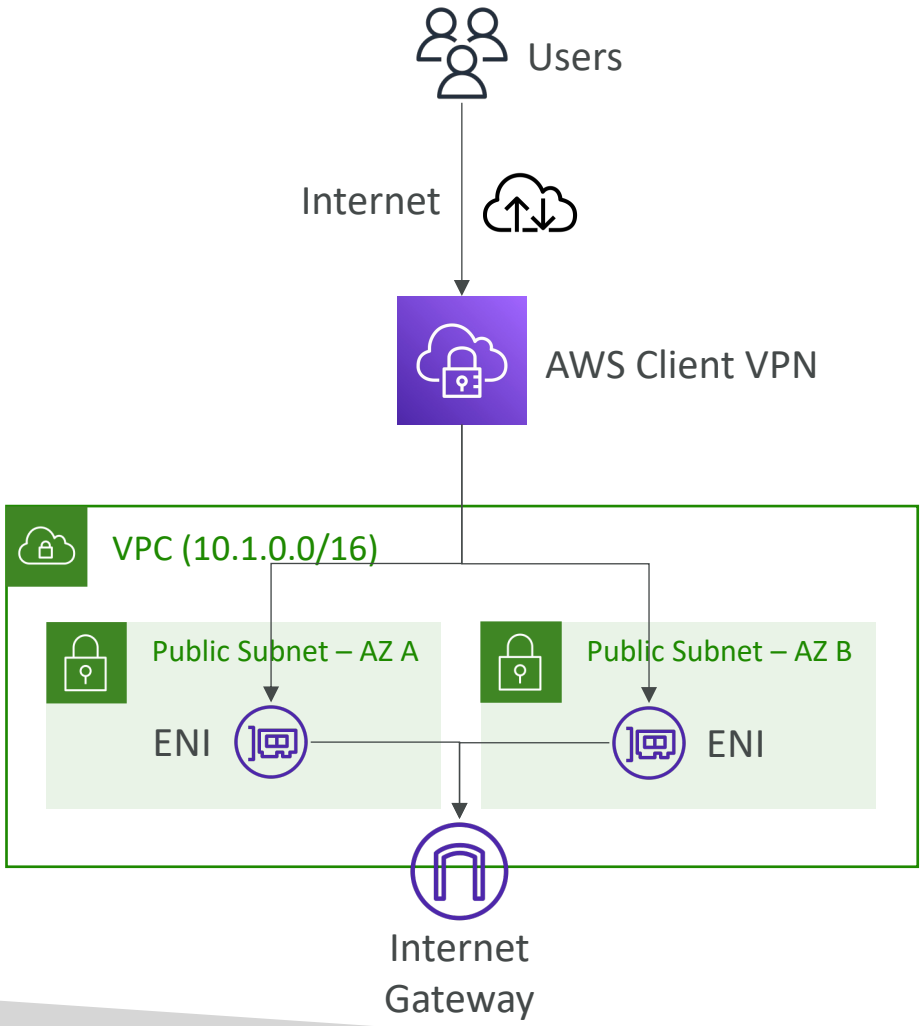
- Access On-Premises resources through AWS with Client VPN

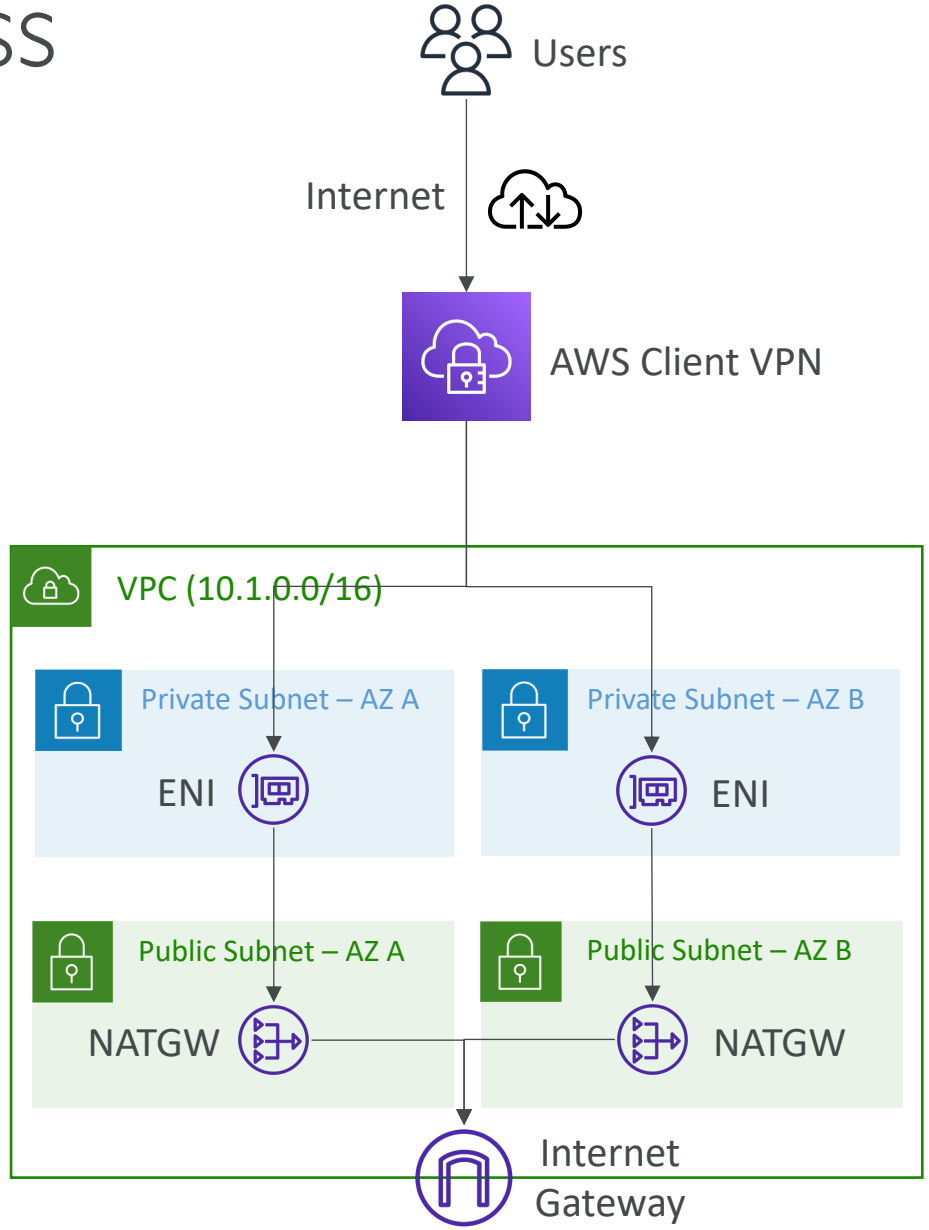


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Client VPN – Internet Access

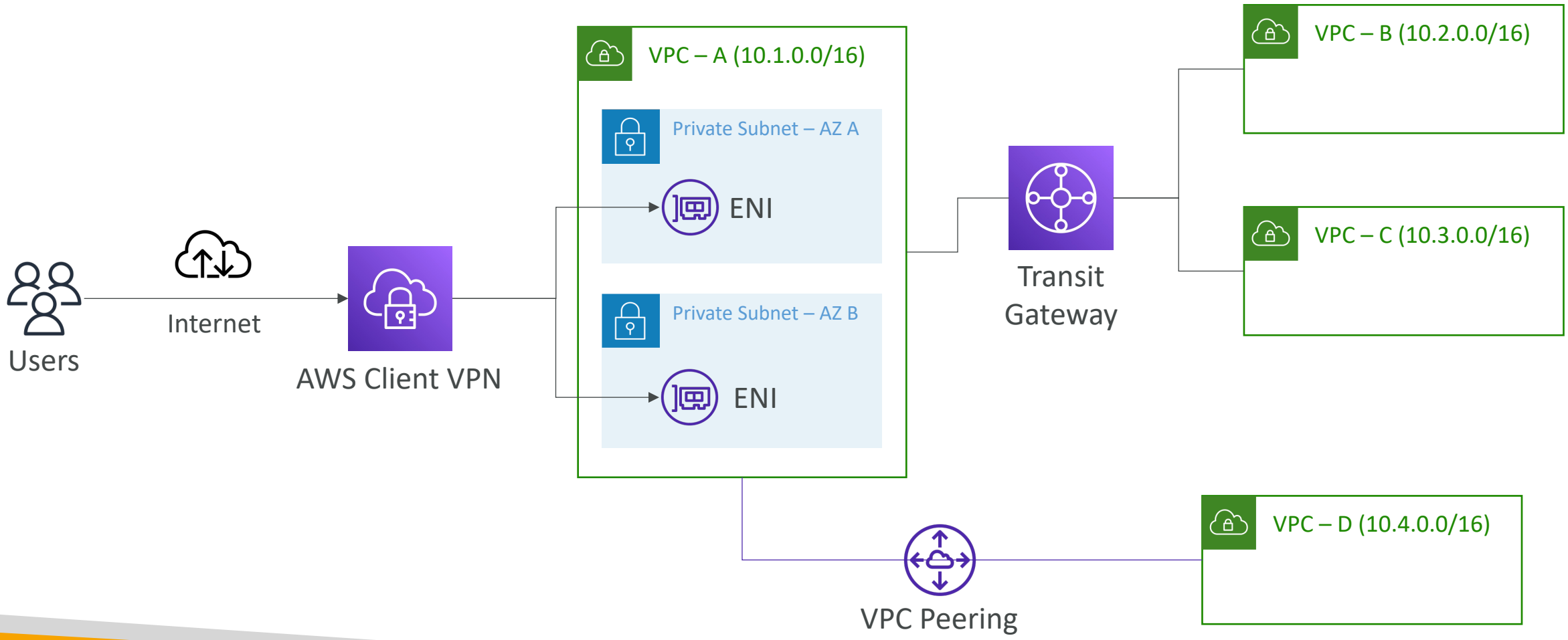




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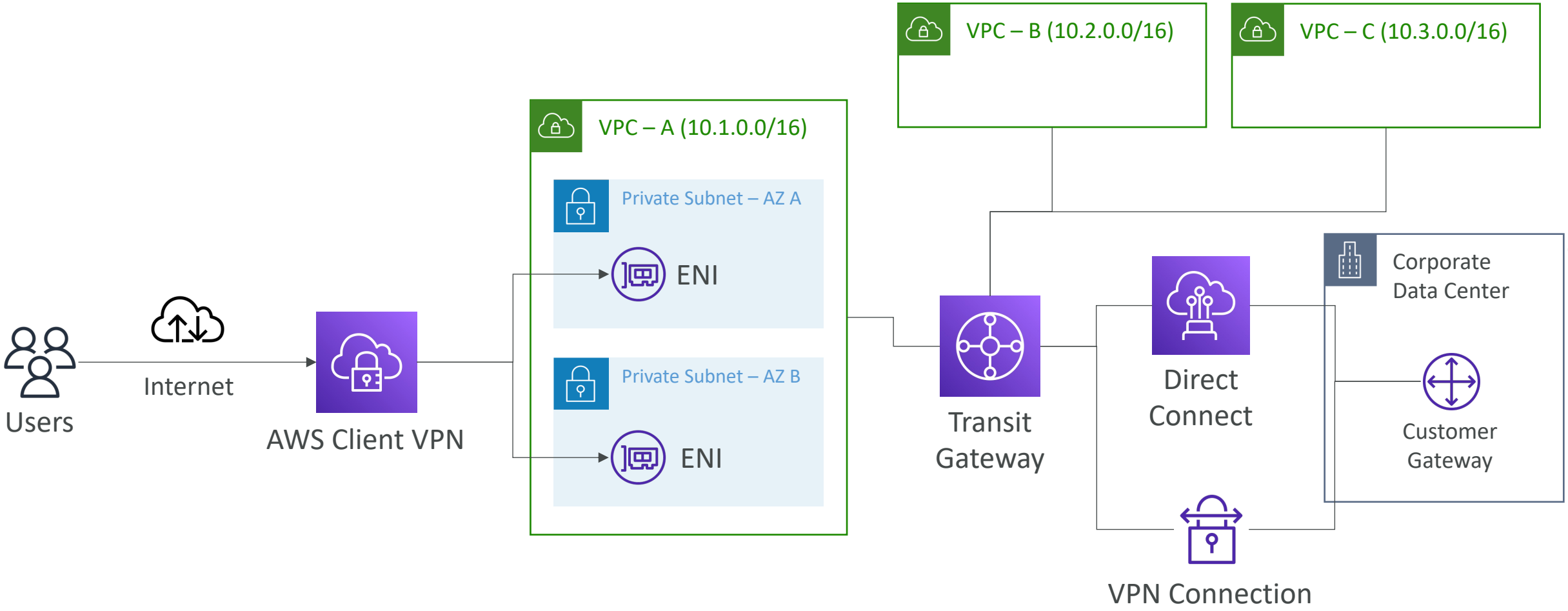
Client VPN – Transit Gateway



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Client VPN – Transit Gateway



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Direct Connect



- Provides a dedicated private connection from a remote network to your VPC
- Dedicated connection must be setup between your DC and AWS Direct Connect locations
- More expensive than running a VPN solution
- Private access to AWS services through VIF
- Bypass ISP, reduce network cost, increase bandwidth and stability
- Not redundant by default (must setup a failover DX or VPN)

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Direct Connect – Virtual Interfaces (VIF)

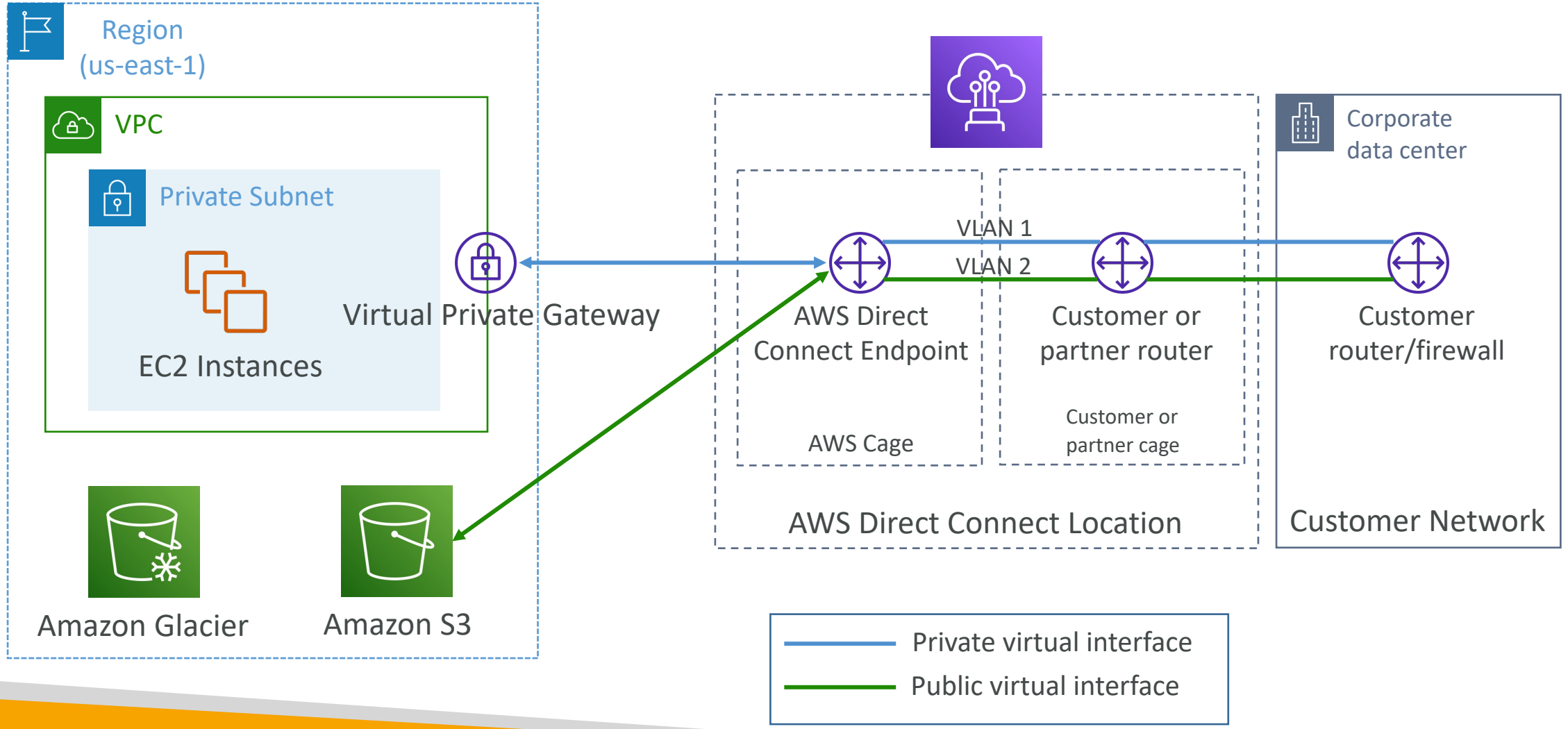
- **Public VIF** – connect to Public AWS Endpoints (S3 buckets, EC2 service, anything AWS …)
- **Private VIF** – connect to resources in your VPC (EC2 instances, ALB, …)
- **Transit Virtual Interface** – connect to resources in a VPC using a Transit Gateway

- VPC Interface Endpoints can be accessed through Private VIF

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Direct Connect Diagram



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Direct Connect – Connection Types

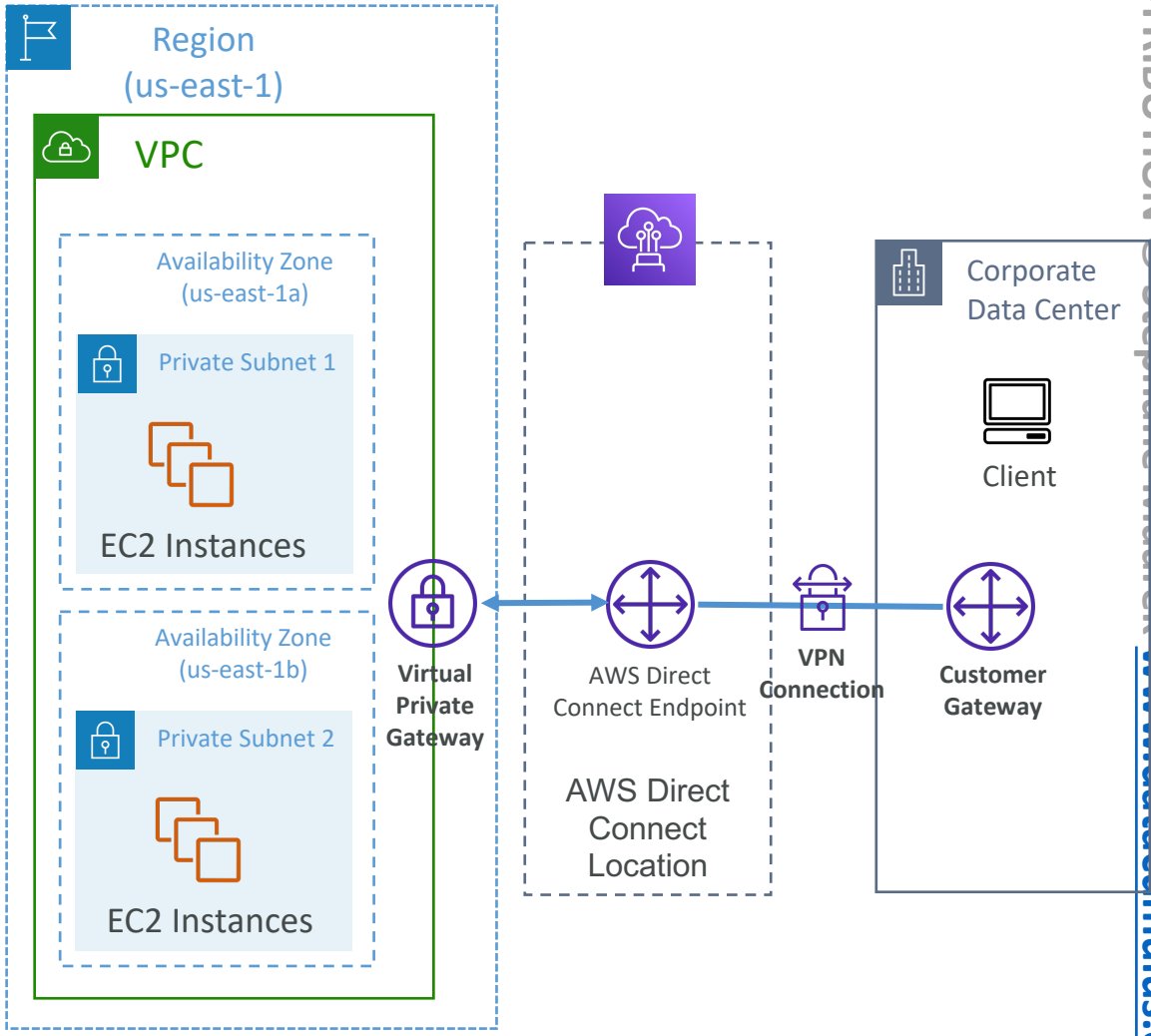
- **Dedicated Connections:** from 1 Gbps to 400 Gbps
  - Physical ethernet port dedicated to a customer
  - Request made to AWS first, then completed by AWS Direct Connect Partners
- **Hosted Connections:** from 50Mbps to 25 Gbps
  - Connection requests are made via AWS Direct Connect Partners
  - Capacity can be **added or removed on demand**
- Lead times are often longer than 1 month to establish a new connection

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Direct Connect – Encryption

- Data in transit is not encrypted but is private
- AWS Direct Connect + VPN provides an IPsec-encrypted private connection
- VPN over Direct Connect connection Uses Public VIF
- Good for an extra level of security, but slightly more complex to put in place

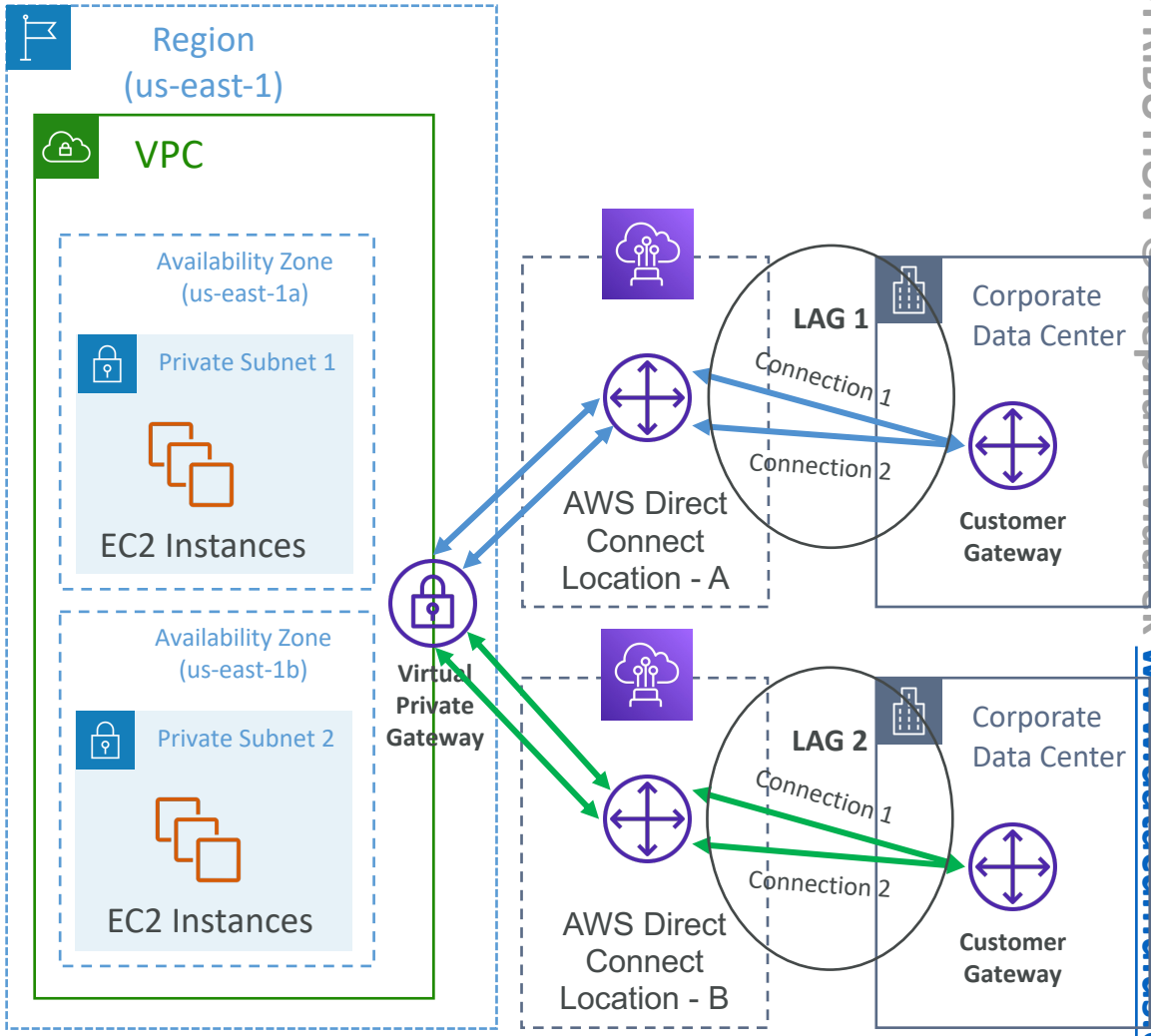


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Direct Connect – Link Aggregation Groups (LAG)

- Get **increased speed** and **failover** by summing up existing DX connections into a **logical one**
- Can aggregate up to 4 connections (active-active mode)
- Can add connections over time to the LAG
- All connections in the LAG:
  - Must be dedicated connections
  - Must have the same bandwidth
  - Must terminate at the same AWS Direct Connect Endpoint
- Can set a minimum number of connections for the LAG to function



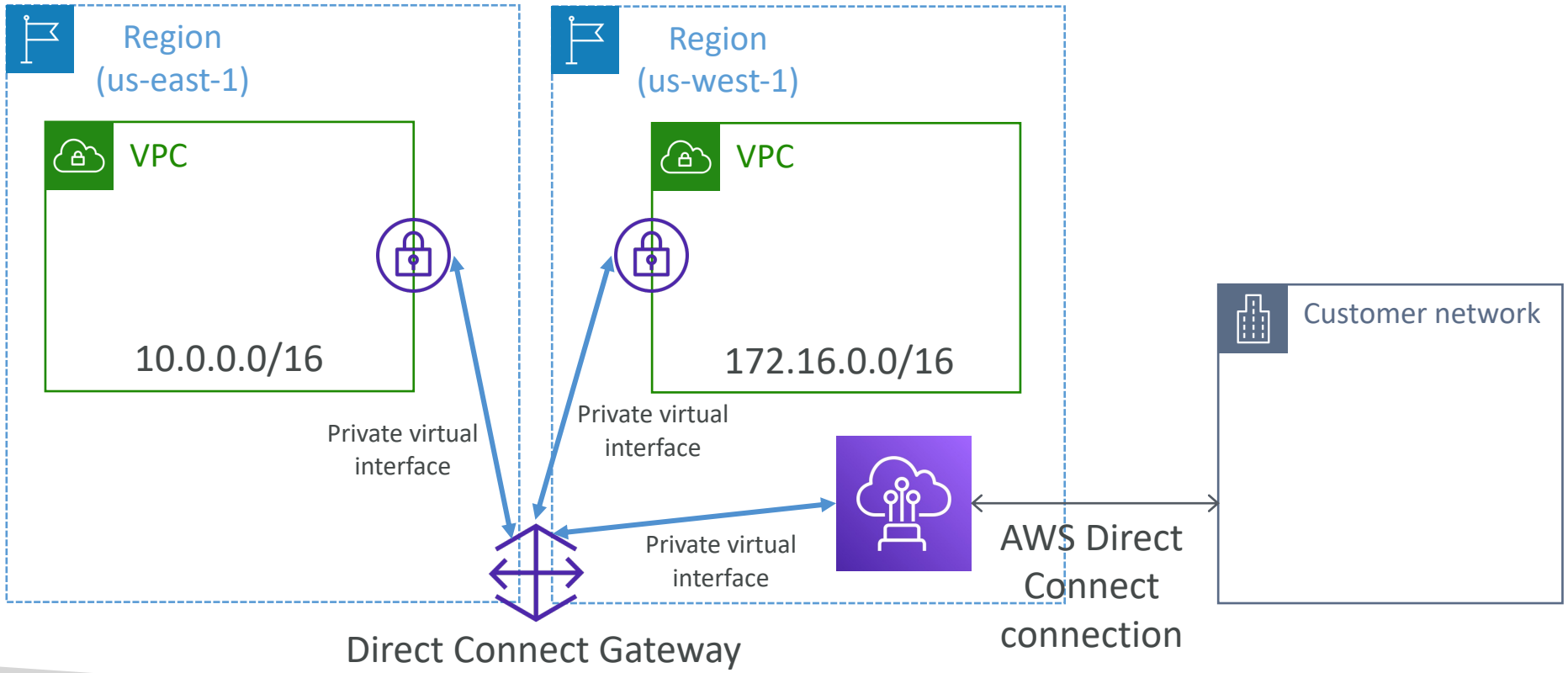
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Direct Connect Gateway



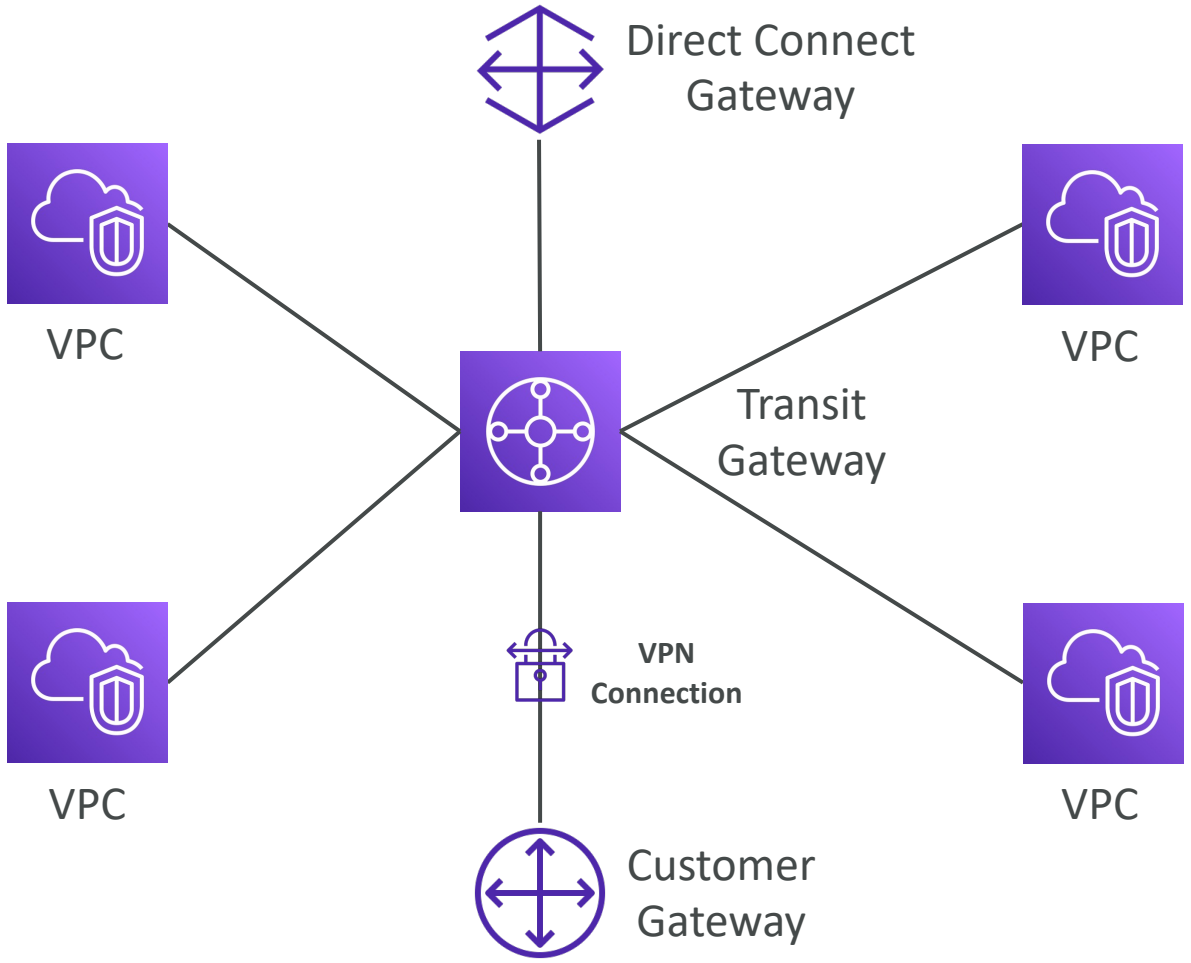
- If you want to setup a Direct Connect to one or more VPC in many different regions (same/cross account), you must use a Direct Connect Gateway



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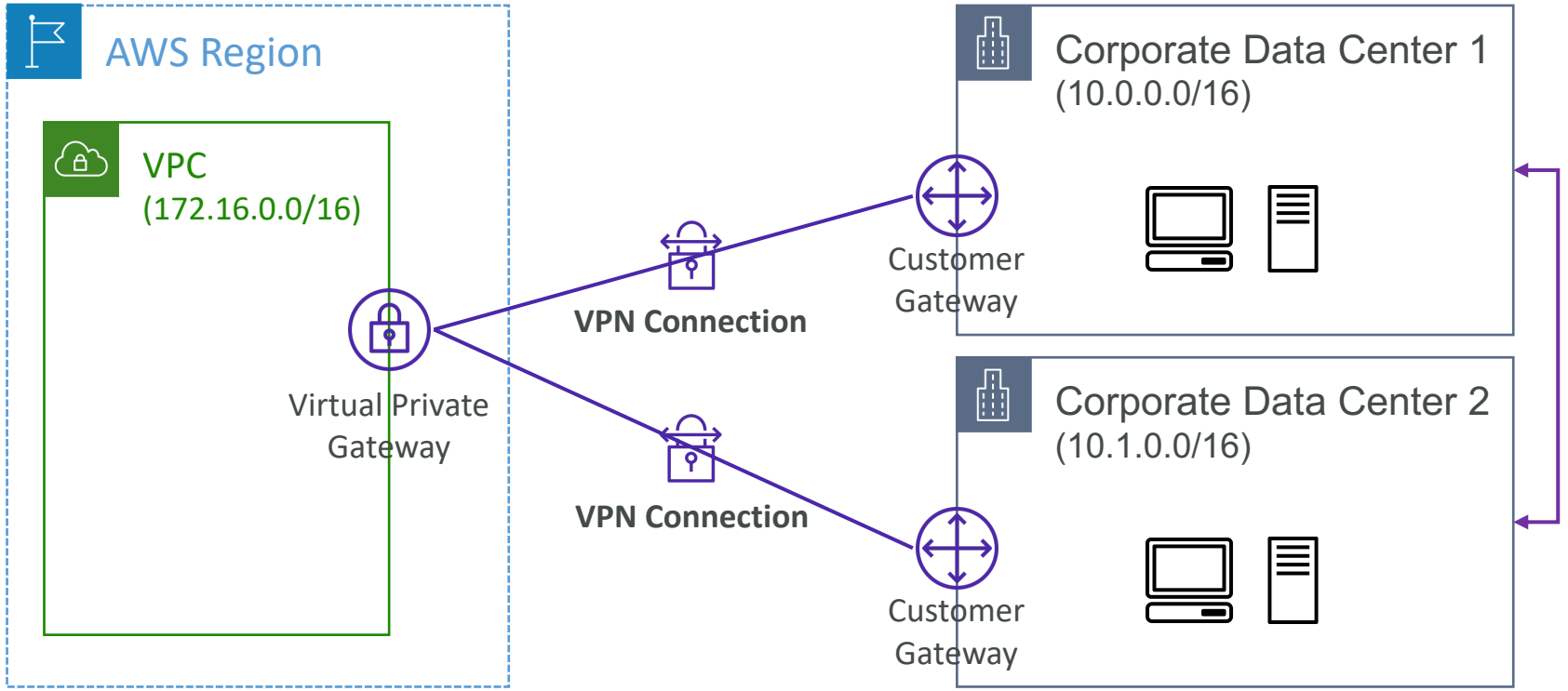
Direct Connect Gateway + Transit Gateway



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Site-to-Site Active-Active Connection

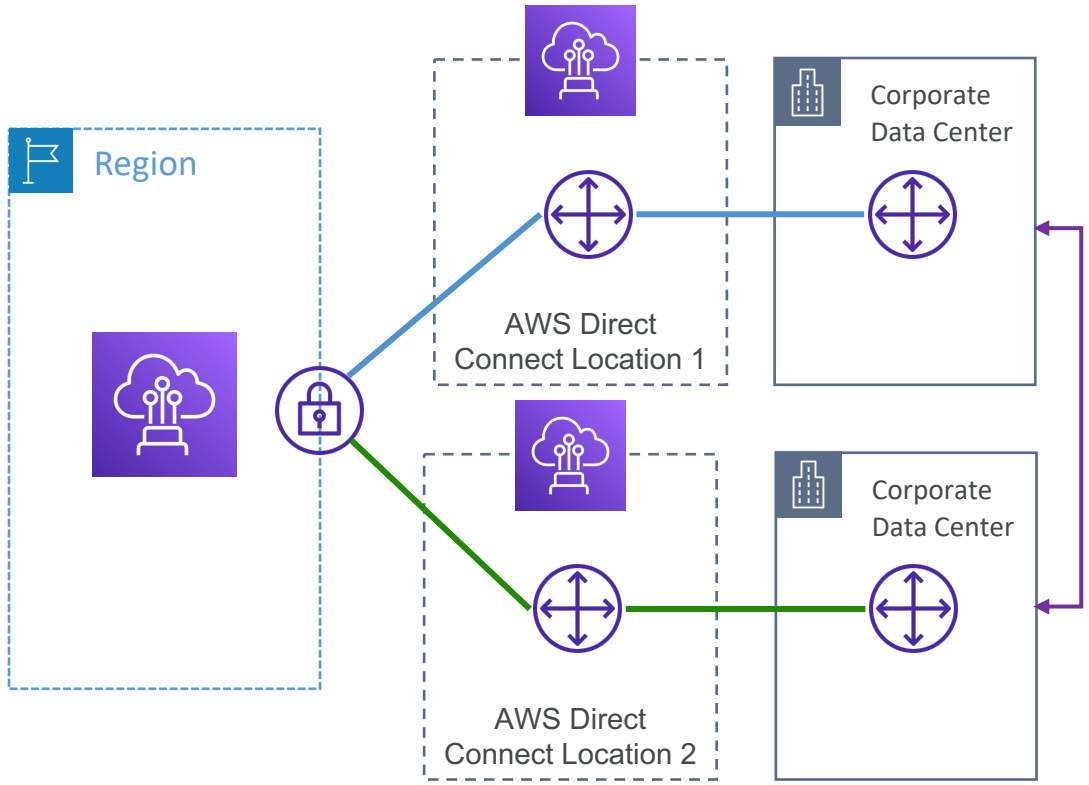


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Direct Connect – High Availability

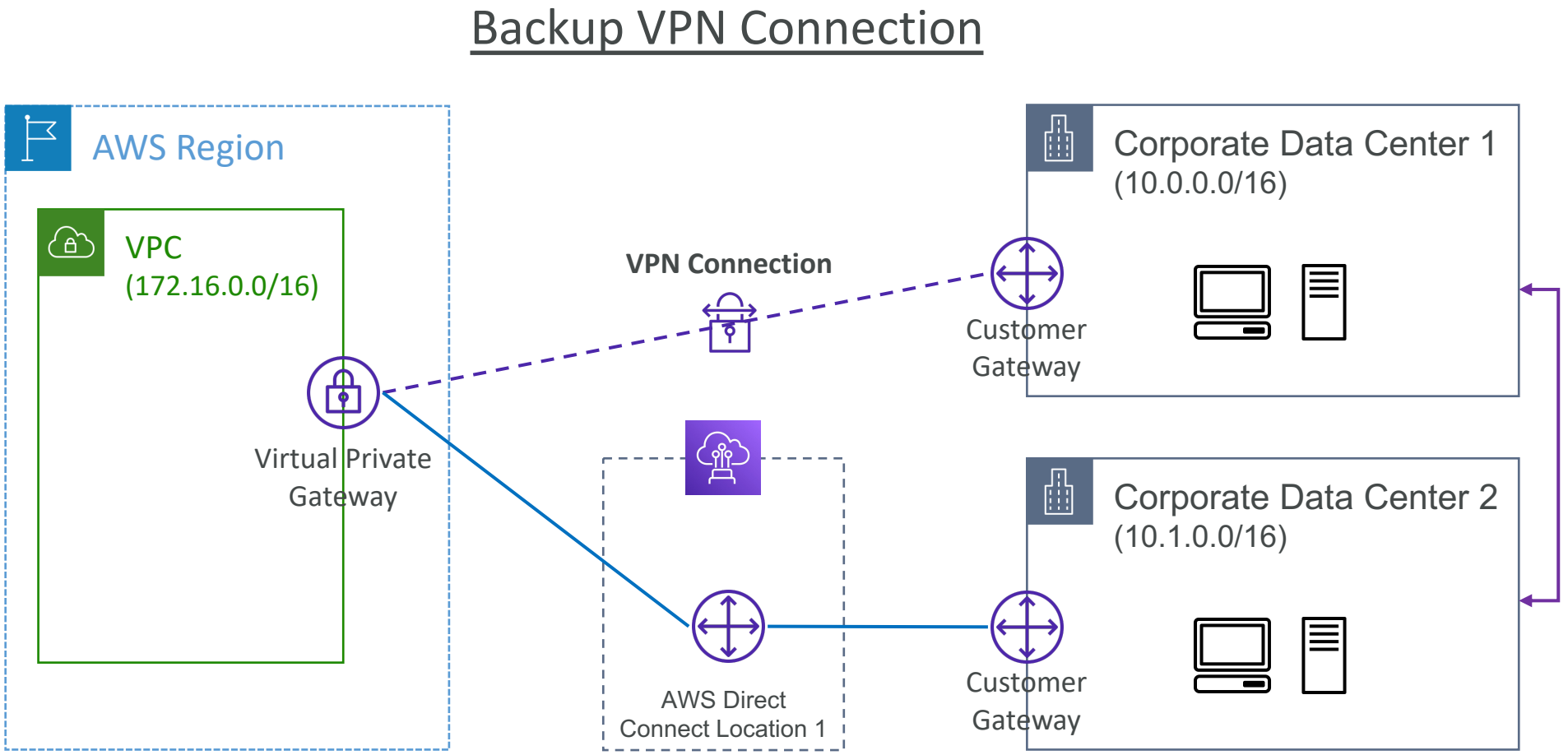
Multiple connections at multiple AWS Direct Connect locations



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Direct Connect – High Availability

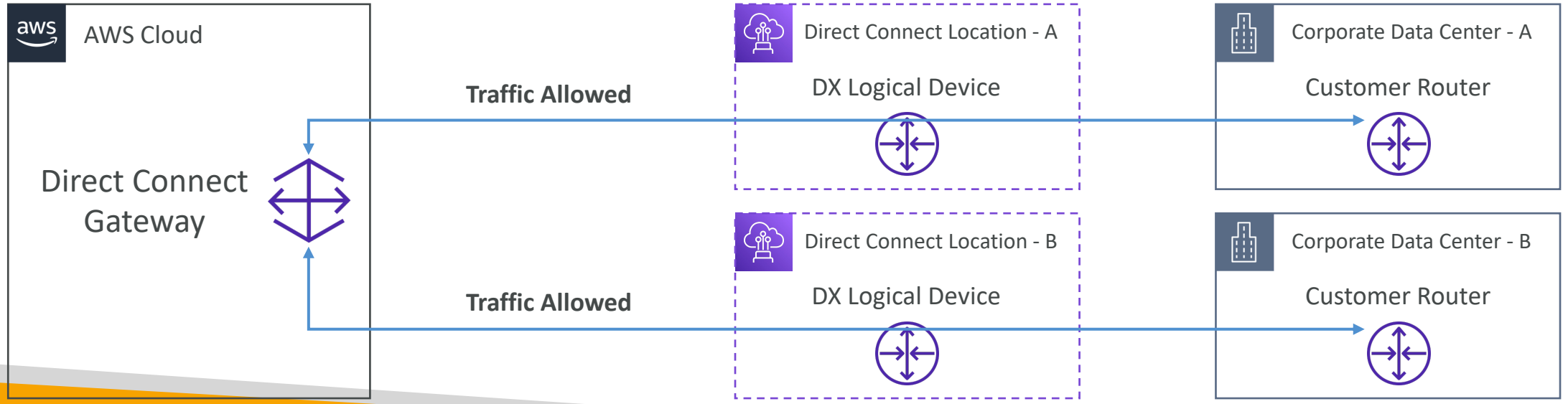


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Direct Connect Gateway – SiteLink

- Allows you to send data from on Direct Connect location to another
- Bypassing AWS Regions
- Use cases: **create private network connections between on-premises data centers by connecting them to Direct Connect locations**
- Data is sent over the fastest path between Direct Connect locations



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VPC Flow Logs

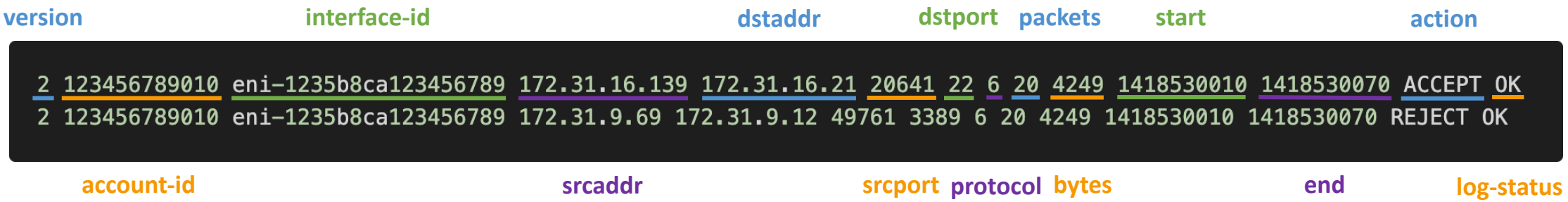


- Capture information about IP traffic going into your interfaces:
  - VPC Flow Logs
  - Subnet Flow Logs
  - Elastic Network Interface (ENI) Flow Logs
- Helps to monitor & troubleshoot connectivity issues
- Flow logs data can go to S3, CloudWatch Logs, and Kinesis Data Firehose
- Captures network information from AWS managed interfaces too: ELB, RDS, ElastiCache, Redshift, WorkSpaces, NATGW, Transit Gateway…

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VPC Flow Logs Syntax



- **srcaddr & dstaddr** – help identify problematic IP
- **srcport & dstport** – help identity problematic ports
- **Action** – success or failure of the request due to Security Group / NACL
- Can be used for analytics on usage patterns, or malicious behavior
- **Query VPC flow logs using Athena on S3 or CloudWatch Logs Insights**
- Flow Logs examples: https://docs.aws.amazon.com/vpc/latest/userguide/flow-logs-records-examples.html

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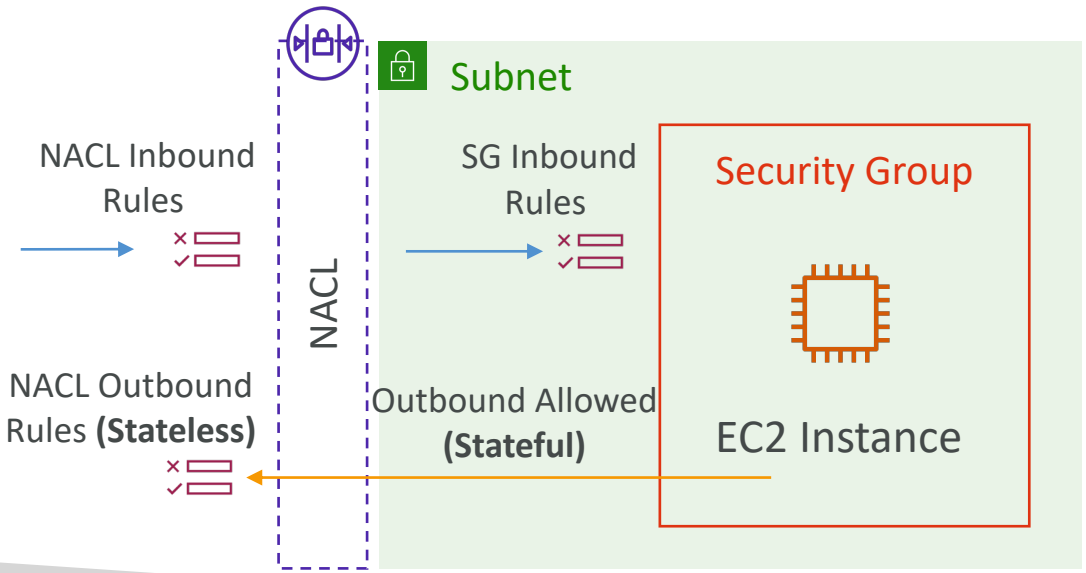
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VPC Flow Logs – Troubleshoot SG & NACL issues

Look at the **"ACTION"** field

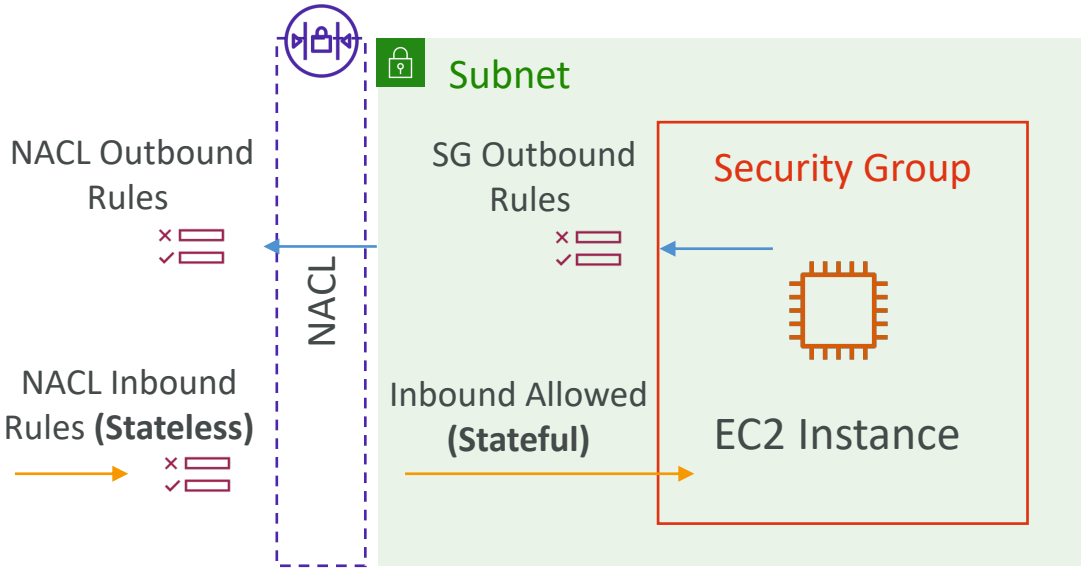
Incoming Requests

- Inbound REJECT => NACL or SG
- Inbound ACCEPT, Outbound REJECT => NACL



Outgoing Requests

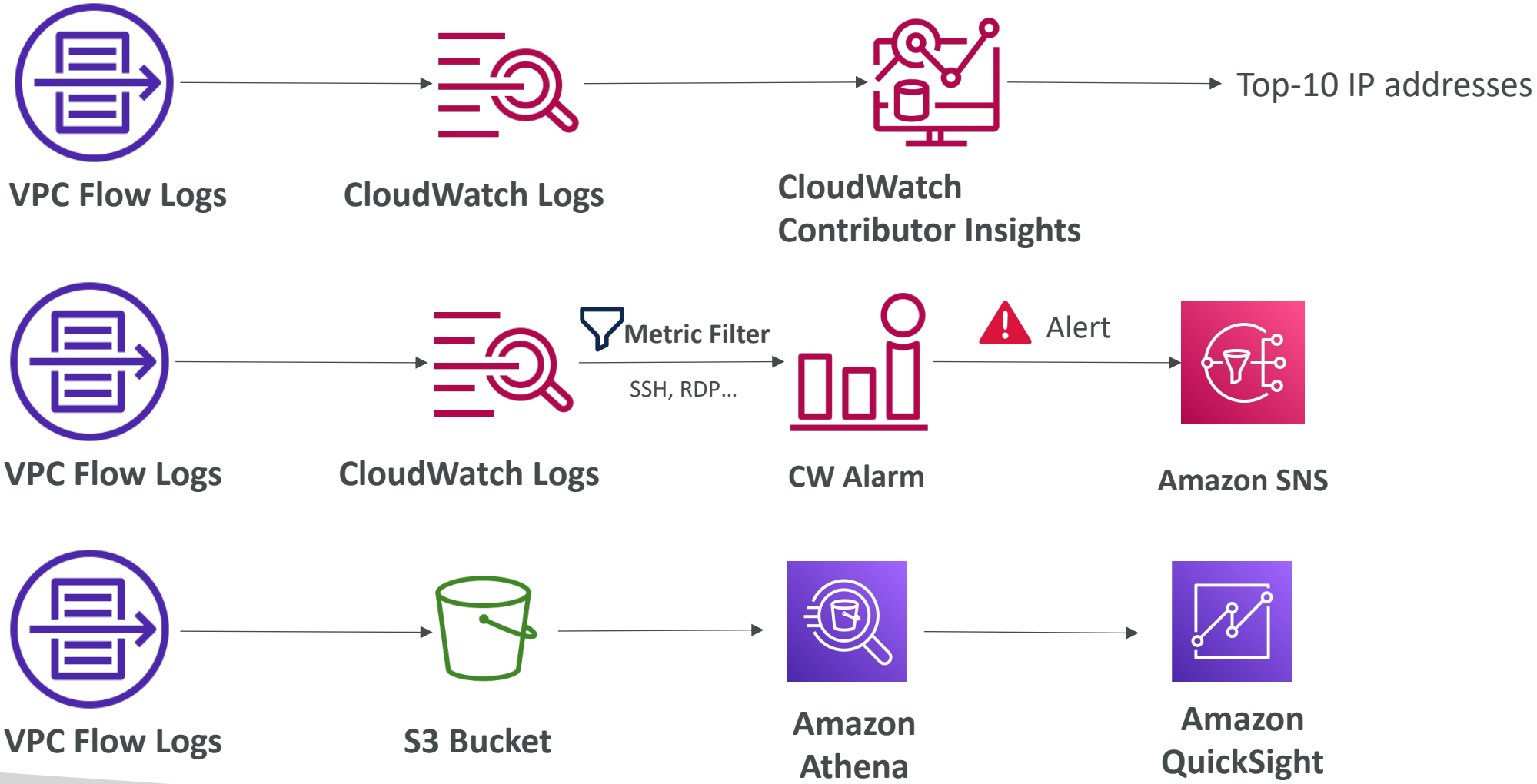
- Outbound REJECT => NACL or SG
- Outbound ACCEPT, Inbound REJECT => NACL



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VPC Flow Logs – Architectures



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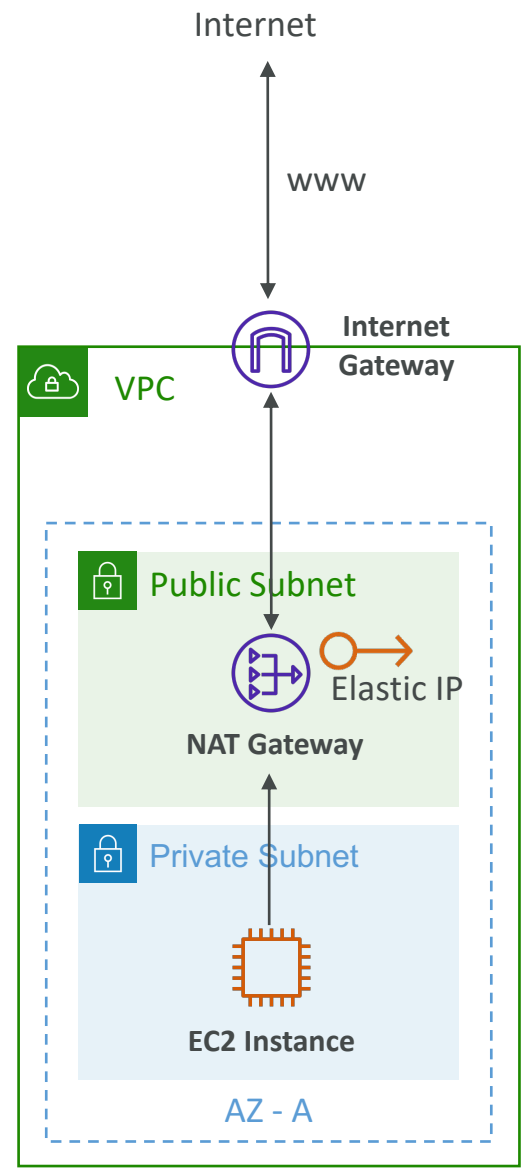
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VPC Flow Logs – with NAT Gateway

- *My Virtual Private Cloud (VPC) flow logs show Action = ACCEPT for inbound traffic coming from public IP addresses. However, my understanding of network address translation (NAT) gateways was that they don't accept traffic from the internet. Is my NAT gateway accepting inbound traffic from the internet?*
- Inbound traffic is permitted by Security Group or NACLs
  - Traffis isn't permitted by the NAT Gateway, it's dropped
  - To confirm run the following query in CloudWatch Log Group

```
filter (dstAddr like 'xxx.xxx' and srcAddr like 'public IP')
| stats sum(bytes) as bytesTransferred by srcAddr, dstAddr
| limit 10
```

- Make 'xxx.xxx' the first two octets of your VPC CIDR
- Replace Public IP with the IP you see in logs
- You will see traffic on the Private IP of the NAT Gateway but nowhere else: traffic was unsolicited and then dropped



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Network Protection on AWS

- To protect network on AWS, we've seen
  - Network Access Control Lists (NACLs)
  - Amazon VPC security groups
  - AWS WAF (protect against malicious requests)
  - AWS Shield & AWS Shield Advanced
  - AWS Firewall Manager (to manage them across accounts)

- But what if we want to protect in a sophisticated way our entire VPC?

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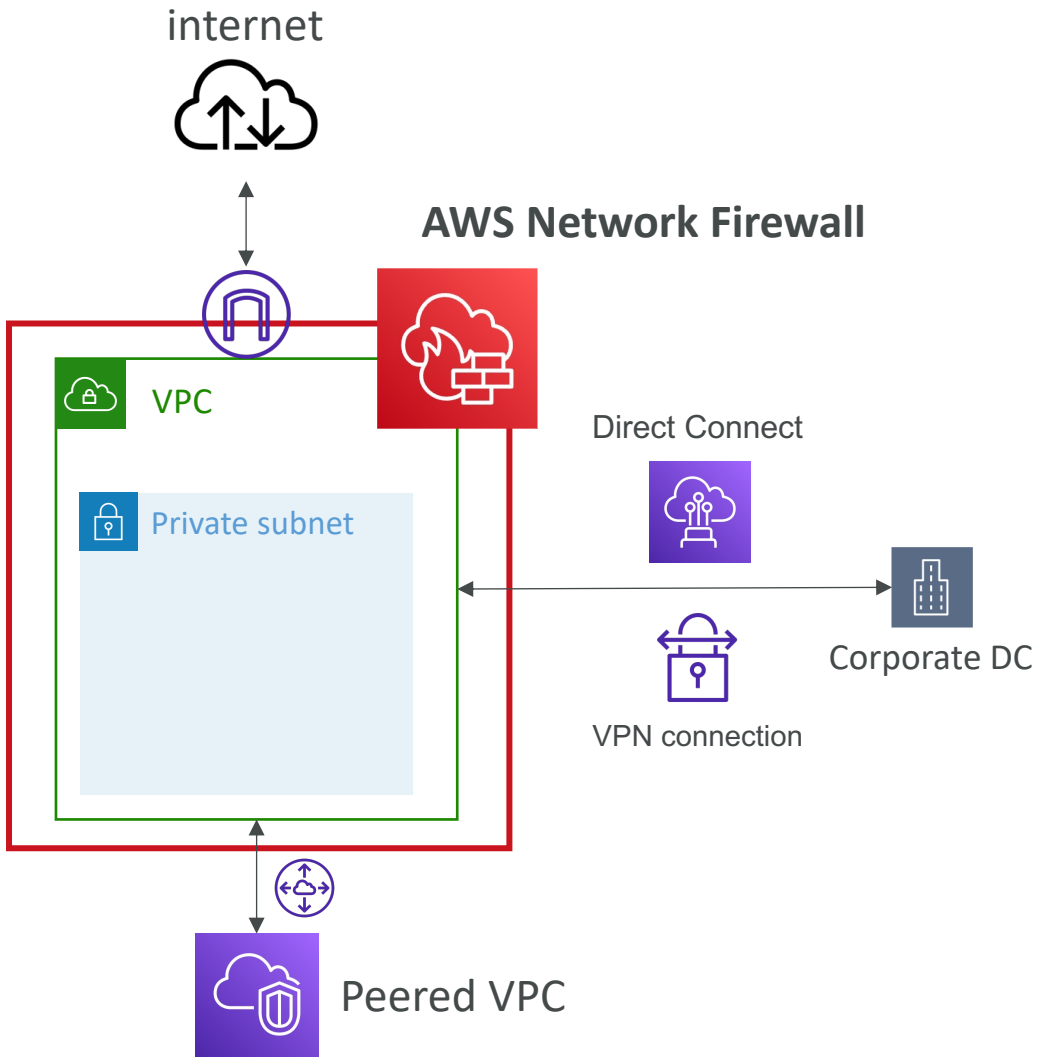
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AWS Network Firewall

- Protect your entire Amazon VPC

- From Layer 3 to Layer 7 protection
- Any direction, you can inspect
  - VPC to VPC traffic
  - Outbound to internet
  - Inbound from internet
  - To / from Direct Connect & Site-to-Site VPN

- Internally, the AWS Network Firewall uses the AWS Gateway Load Balancer
- Rules can be centrally managed cross-account by AWS Firewall Manager to apply to many VPCs



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Network Firewall – Fine Grained Controls



- Supports 1000s of rules
  - IP & port - example: 10,000s of IPs filtering
  - Protocol – example: block the SMB protocol for outbound communications
  - Stateful domain list rule groups: only allow outbound traffic to *.mycorp.com or third-party software repo
  - General pattern matching using regex
- **Traffic filtering: Allow, drop, or alert for the traffic that matches the rules**
- **Active flow inspection** to protect against network threats with intrusion-prevention capabilities (like Gateway Load Balancer, but all managed by AWS)

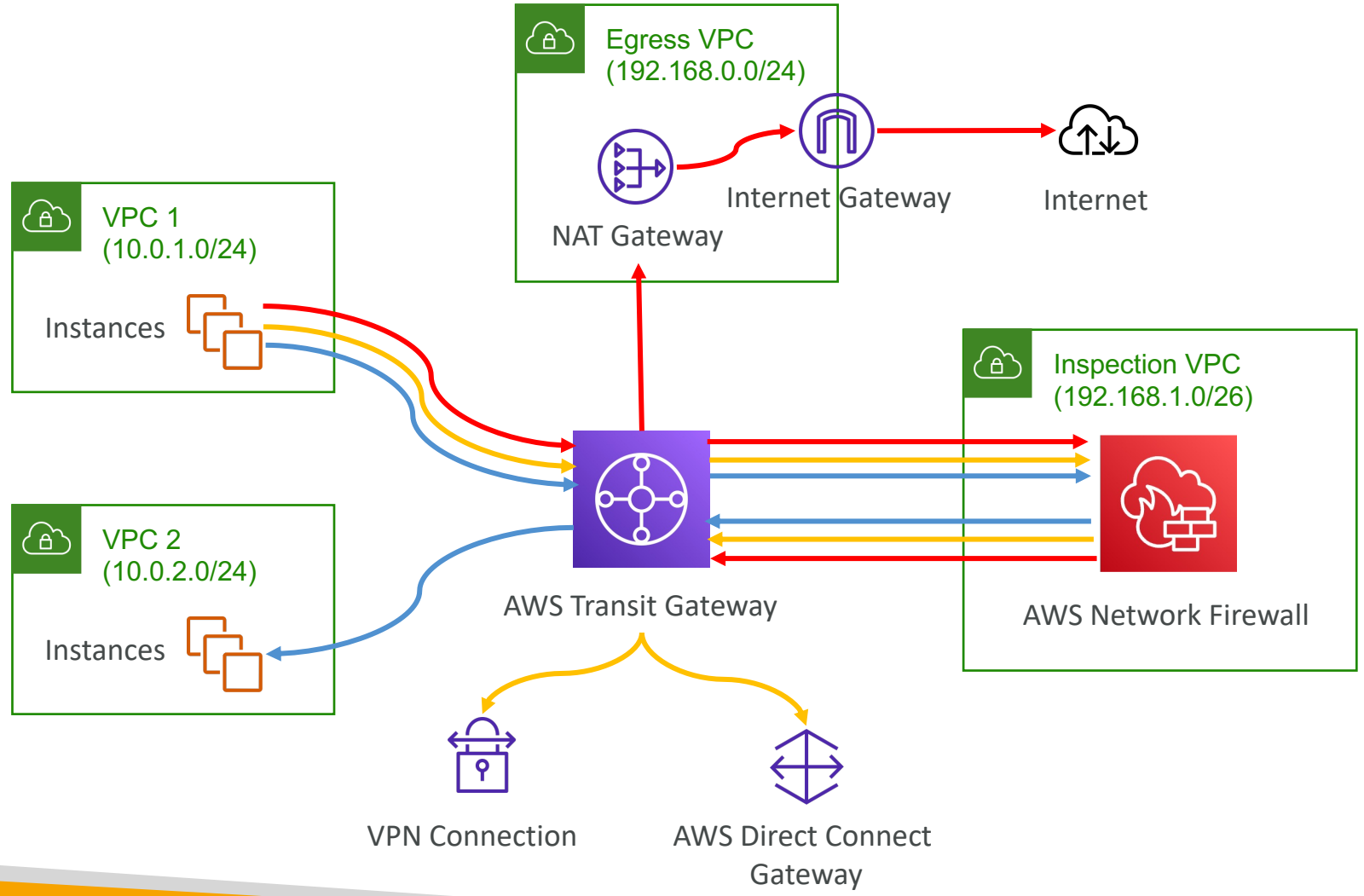
- Send logs of rule matches to Amazon S3, CloudWatch Logs, Kinesis Data Firehose

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AWS Network Firewall - Architecture

- North-South VPC egress to Internet
- North-South VPC to VPN or Direct Connect
- East-West VPC to VPC



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